

Doubly robust conditional independence testing with generative neural networks

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Abstract

This article addresses the problem of testing the conditional independence of two generic random vectors X and Y given a third random vector Z , which plays an important role in statistical and machine learning applications. We propose a new non-parametric testing procedure that avoids explicitly estimating any conditional distributions but instead requires sampling from the two marginal conditional distributions of X given Z and Y given Z . We further propose using a generative neural network (GNN) framework to sample from these approximated marginal conditional distributions, which tends to mitigate the curse of dimensionality due to its adaptivity to any low-dimensional structures and smoothness underlying the data. Theoretically, our test statistic is shown to enjoy a double robustness property against GNN approximation errors, meaning that the test statistic retains all desirable properties of the oracle test statistic utilizing the true marginal conditional distributions, as long as the product of the two approximation errors decays to zero faster than the parametric rate. Asymptotic properties of our statistic and the consistency of a bootstrap procedure are derived under both null and local alternatives. Extensive numerical experiments and real data analysis illustrate the effectiveness and broad applicability of our proposed test.

Keywords: conditional distribution, double robustness, generative models, Kernel method, maximum mean discrepancy

1 Introduction

Conditional independence (CI) testing plays an important role in various areas of statistics and machine learning. For example, it can be used to examine causal relations (Pearl, 2009), determine the structure of graphical models (Koller & Friedman, 2009) and simplify statistical models by performing variable selection (Azadkia & Chatterjee, 2021; George, 2000) and dimension reduction (Cook & Li, 2002; B. Li, 2018). In this article, we focus on testing if two generic random vectors $X \in \mathbb{R}^{d_X}$ and $Y \in \mathbb{R}^{d_Y}$ are conditionally independent given a third random vector $Z \in \mathbb{R}^{d_Z}$, where d_X , d_Y , and d_Z are positive integers; that is, we test the null hypothesis $H_0 : X \perp\!\!\!\perp Y \mid Z$ against $H_1 : X \not\perp\!\!\!\perp Y \mid Z$. In general, CI testing is statistically more challenging than independence testing because it requires comparing the joint conditional distribution with the product of two of its marginals simultaneously for each value realized by the conditioning variable Z . For example, as shown in Shah and Peters (2020), there does not exist a CI test that can control the type-I error for all joint distributions of (X, Y, Z) under H_0 while having nontrivial power against any alternative. Typically, to make the conditional independence testing problem feasible, one must restrict the distribution class by introducing identifiable structures to make conditional distributions statistically estimable; see, for example, Neykov et al. (2021). A common identifiable structure employed in classic literature on conditional distribution estimation (e.g. Bashtannyk & Hyndman, 2001; Izbicki & Lee, 2016; M. Li et al., 2022) involves imposing smoothness

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conditions on the conditional densities of $X, Y|Z = z$ as z varies across its support. However, when Z represents a random high-dimensional complex object, such as images or texts, the conditional distributions of $X, Y|Z = z$ may not possess a density function; additionally, a smoothness condition alone may be insufficient to capture many low-dimensional intrinsic structures underlying the true conditional distributions.

Motivated by the high-dimensional nature of data in applications such as cancer genetics and machine learning (e.g. image generation and classification), this article focuses on scenarios where Z has a large dimension, and one or both of X and Y are high-dimensional. Such data structures are also prevalent in computer vision tasks, including image denoising, deblurring, object detection, and semantic segmentation. For example, X and Y can be the noisy and clear images of the same object, and Z corresponds to the feature maps of an encoder-decoder network trained for an image denoising task. Our proposed test can be used to determine the dimension and structure of the feature map Z , which will provide guidance on network fine-tuning; see Section 5 for related real-data applications.

1.1 Related literature

The literature on non-parametric conditional independence (CI) testing is extensive, with a recent survey provided by [C. Li and Fan \(2020\)](#). The main challenges in CI testing emerge in high-dimensional settings, where the goal is to construct a test that maintains strong empirical performance, accurate asymptotic size, and high power against any alternative hypothesis. To the best of our knowledge, existing tests typically prioritize one or two of these features, making it challenging to develop a test that simultaneously excels in all three aspects. Below, we provide a point-by-point discussion of how these challenges arise in high-dimensional settings and how existing CI tests have addressed them.

1.1.1 Achieving good performance in high-dimensional setting

This challenge in CI testing arises when attempting to construct a sample version of a population-level CI measure, which often involves conditional expectations such as $\mathbb{E}[u(X)|Z]$ and/or $\mathbb{E}[v(Y)|Z]$ for some real or function valued mappings u and v . Examples include CI testing based on conditional characteristic functions ([Su & White, 2007](#); [Wang et al., 2015](#)), where $u(x) = e^{itx}$ for all $t \in \mathbb{R}$ and conditional cumulative distribution functions ([Cai et al., 2022](#); [Su & White, 2014](#)), where $u(x) = 1(x \leq t)$ for all $t \in \mathbb{R}$ with $1(\cdot)$ denoting the indicator function. Direct estimation of these conditional expectations can be inefficient and cumbersome, often requiring kernel smoothing and numerical integration, which can be difficult to implement and inaccurate in high-dimensional settings. The need for explicit estimation of conditional expectations exposes existing methods to the curse of dimensionality: their performance significantly deteriorates as the dimension d_Z of Z and/or the dimensions (d_X, d_Y) of (X, Y) become larger ([X. Zhou et al., 2022](#), Section 1).

Another approach to CI testing, when the limiting null distribution of the test statistic is not pivotal, involves conditional permutation. For instance, in ([Fukumizu et al., 2007](#); [Kim et al., 2022](#); [Neykov et al., 2021](#)), permutations are performed locally on bins or clusters based on the similarity of the conditioning variable Z . However, this local permutation procedure requires a large sample size and may become unreliable when Z is a continuous random vector or when the dimension of Z is large ([Z. Huang et al., 2022](#), Section 1). Alternatively, [Sen et al. \(2017\)](#) and [S. Li et al. \(2023\)](#) proposed using one-nearest neighbour sampling to perform local permutation. Nevertheless, their approach requires sample splitting and imposes strong smoothness assumptions which can be difficult to verify in practice.

From a different perspective, the challenge discussed above can be attributed to the difficulty of sampling from the conditional distribution of X and/or Y given Z , since in practice we only have i.i.d. copies of (X, Y, Z) and the true conditional distributions are unknown. If we had access to a (conditional) generative model for sampling from these conditional distributions, we could easily approximate the conditional expectations—crucial components of most CI test procedures—to arbitrary accuracy, using Monte Carlo methods without sacrificing computational efficiency. Recently, generative neural network (GNN)-based approaches, such as generative adversarial networks (GAN) ([Goodfellow et al., 2014](#)) and generative moment matching networks (GMMN)

(Dziugaite et al., 2015; Y. Li et al., 2015), have seen great success in generating high-quality samples, including images and text, from complex and potentially high-dimensional distributions by passing random noise through a deep neural network learned from training data. Along with their impressive empirical performance, there has been a surge in theoretical works that examine the properties of GNNs. For example, Tang and Yang (2023) derived the minimax rate for implicitly estimating a high-dimensional distribution residing on a low-dimensional manifold within a generative model framework, showing that the rate depends on the manifold's intrinsic dimension and smoothness, as well as the smoothness of the probability density. Additionally, Chen et al. (2020) demonstrated that GANs are consistent estimators of distributions with certain smooth densities or low-dimensional structures.

Recent works such as Shi et al. (2021) and Bellot and der Schaar (2019) utilize GANs to approximate and sample from conditional distributions as part of their CI testing procedures. The double GAN-based conditional independence test (DGCIT) proposed in Shi et al. (2021) and the generative conditional independence test (GCIT) proposed in Bellot and der Schaar (2019) both demonstrate satisfactory performance in high-dimensional settings. However, DGCIT struggles with power against certain alternative hypotheses, and GCIT requires strong assumptions to ensure size accuracy, which may not always be feasible in practice. More details on these methods can be found in Appendix A.1, online supplementary material.

1.1.2 Achieving accurate asymptotic size

Due to the non-parametric nature of (conditional) expectation estimation, the estimation error of nuisance parameters is often non-negligible, making it challenging to construct a CI test with theoretical guarantees on size accuracy. One approach to circumvent this challenge is by imposing additional assumptions on the distribution of (X, Y, Z) . For instance, the model-X assumption in Candès et al. (2018) posits that the conditional distribution of X given Z is known. However, the validity of such tests heavily depends on the model-X assumption, which may be too restrictive in practice.

In the fully non-parametric setting, accurate asymptotic size can also be achieved if the estimation errors have an explicit form and can be handled directly. For example, the test proposed in Banerjee et al. (2024) and Wang et al. (2015) use kernel smoothing to estimate the conditional expectations and their statistics are scaled by nb^r , where $b \rightarrow 0$ and $r > 0$ are parameters involved in kernel smoothing. While these tests can achieve accurate asymptotic size, they may experience substantial power loss due to the smaller scaling factor, and will suffer from the curse of dimensionality. Furthermore, the choice of b will also impact the finite sample size accuracy.

Another approach to achieve accurate asymptotic size is through achieving double robustness. Typically, a doubly robust CI statistic is a sample version of $\mathbb{E}[(u(X) - \mathbb{E}[u(X)|Z])(v(X) - \mathbb{E}[v(X)|Z])]$, which is the product of residuals after regressing $u(X)$ and $v(Y)$ on Z for some (class of) functions u and v . In this case, the decay rate of the difference between the statistic and its oracle version depends on the product of the estimation errors of $\mathbb{E}[u(X)|Z]$ and $\mathbb{E}[v(Y)|Z]$, and this product is assumed to decay faster than the $n^{-1/2}$ parametric rate, which is sufficient to guarantee accurate asymptotic size. Examples include the CI tests proposed in Shah and Peters (2020) and Niu et al. (2024) and DGCIT. The main issue with this type of statistic is that it may fail to detect certain alternative hypotheses, as $\mathbb{E}[(u(X) - \mathbb{E}[u(X)|Z])(v(X) - \mathbb{E}[v(X)|Z])]$ cannot fully characterize conditional independence; see Appendix A.2, online supplementary material for more details. That being said, there do exist doubly robust CI tests that are powerful against any alternative hypothesis, such as the CIT test proposed in Cai et al. (2022). However, their method involves estimating the conditional cumulative distribution functions (CDFs) of X and Y given Z , which suffers from the curse of dimensionality; see Section 4.2 and Appendix A.2, online supplementary material.

1.1.3 Maintaining power against any alternative hypothesis

For a CI test to maintain power against any alternative hypothesis, the population CI measure, typically a functional of the joint distribution of (X, Y, Z) , must fully characterize conditional independence. These tests are generally more complex and involve estimating real or Hilbert space-valued functions of Z . One family of such tests are metric-based CI tests (C. Li & Fan, 2020).

For example, the tests proposed in Wang et al. (2015) and Cai et al. (2022) are based on characteristic functions, and the test proposed in Banerjee et al. (2024) is based on the conditional version of ball divergence (Pan et al., 2018). For a detailed account of tests in this family, we refer to Section 3.1.2 of C. Li and Fan (2020).

Another popular and important family of CI tests that is powerful against any alternative is kernel-based CI tests (Fukumizu et al., 2007; Z. Huang et al., 2022; Pogodin et al., 2022, 2024; Scetbon et al., 2022; K. Zhang et al., 2011). The population CI measure used by these tests compares the (kernel) mean embeddings of distributions into some reproducing kernel Hilbert space (RKHS), and under mild assumptions on the kernel functions, the CI measure can fully characterize conditional independence. One class of kernel-based CI tests includes tests based on the Hilbert–Schmidt conditional independence criterion (HSCIC), which compares the mean embedding of the conditional distribution of (X, Y) given Z with the product of mean embeddings of X and Y given Z . As a direct generalization of the Hilbert–Schmidt independence criterion (Gretton et al., 2005) used in independence testing, HSCIC is used in Sheng and Sriperumbudur (2023) and Park and Muandet (2020) to construct CI tests. However, no theoretical results on the size accuracy of these tests are derived; see Appendix A.3, online supplementary material.

Another class of kernel-based CI tests is not straightforward generalizations from unconditional independence testing. The population CI measure used by tests in this class involves the conditional cross-covariance operator. Examples include the tests proposed in Fukumizu et al. (2007), K. Zhang et al. (2011), Pogodin et al. (2024), and Pogodin et al. (2022). One major issue faced by tests in this class is the slow convergence rate for conditional mean embedding estimators, which may cause large size distortion and power loss when dimension of Z is large; see Appendix A.3 in the online supplementary material for more details.

1.2 Our contributions

In this article, we propose a GNN based CI test that can address all above-mentioned challenges. In particular, the proposed test exhibits promising empirical performance when the dimension of Z is large, is doubly robust against approximation errors of conditional distributions and remains powerful under all alternatives in H_1 . In a nutshell, the proposed test is an empirical version of a population-level CI measure based on the maximum mean discrepancy (MMD, Gretton et al., 2012) between the distribution of (X, Y, Z) under H_0 and its true distribution; as a result, it fully characterizes CI under mild assumptions on the employed kernel functions. In addition, we implicitly estimate the mean embeddings of (marginal) conditional distributions via sampling from GNNs trained through sample splitting and cross-fitting. Since the training loss of the GNNs is closely related to the relevant metric in the proposed test statistic, no explicit requirements are needed on the dimensions of X , Y , and Z to guarantee size control and good power of the testing procedure, provided the trained GNNs are sufficiently accurate; see Section 2.1 and Remark 3 of Section 3. Due to the careful incorporation of the GNNs, our test statistic is shown to enjoy a double robustness property against GNN approximation errors, meaning that the test statistic will retain all desirable properties of the oracle test statistic utilizing the true marginal conditional distributions, as long as the product of the two approximation errors decays to zero faster than the parametric rate. This double robustness property allows more flexibility and tolerance in estimating the two marginal conditional distributions, making our testing procedure less sensitive to the GNN approximation errors, which typically suffer from a slower non-parametric rate. Specifically, asymptotic properties of our statistic, as well as the consistency of a bootstrap procedure, are derived under both null and local alternatives. Finally, extensive numerical experiments and real data analysis illustrate the effectiveness and broad applicability of our proposed test.

1.3 Organization and notations

The rest of this article is organized as follows. In Section 2, we first motivate and propose our test statistic for CI testing with a bootstrap calibration procedure, and formally introduce the notation of double robustness property in the context of CI testing. In Section 3, we develop asymptotic properties of our test statistic under H_0 and local alternatives, as well as the consistency results for the proposed wild bootstrap procedure. Section 4 evaluates and compares our proposed test along with other representative methods in the literature via finite sample simulations.

In Section 5, two real data examples are provided using the MNIST dataset. Section 6 provides some concluding remarks. All the proofs, implementation details and another real data example using Cancer Cell Line Encyclopedia dataset are deferred to the [online supplementary material](#). The code for reproducing the numerical results, additional implementation details, and instructions for accessing the real data used in the article are available at <https://github.com/LinjunHuang86749/MMDCI>.

The following notations will be used throughout the article. For any positive integer d and random vectors $(X^1, X^2, \dots, X^d, Z)$ defined on the same probability space, we use $P_{X^1 X^2 \dots X^d}$ and $P_{X^1 X^2 \dots X^d | Z}$ to denote the joint distribution of $X^1, X^2, \dots, X^d$ and its conditional distribution given Z . Let $\mathbb{E}_Z$ or $\mathbb{E}_{Z \sim Q}$ denote taking expectation with respect to random variable Z with distribution Q and P_m denote the Lebesgue measure on $\mathbb{R}^d$. For positive integer n , define $[n] = \{1, 2, \dots, n\}$. For probability measure $\mu(\cdot)$ on $\mathbb{R}^d$ and $p \geq 1$, let $L_p(\mathbb{R}^d, \mu) = \{f: \mathbb{R}^d \rightarrow \mathbb{R}: \int |f(x)|^p d\mu(x) < \infty\}$. For any Hilbert spaces A and B , let $A \otimes B$ denote their tensor product. Given random vectors $a, b \in \mathbb{R}^d$, the Gaussian kernel is defined as $\mathcal{K}(a, b) = \exp(-\|a - b\|_2^2 / \sigma)$ and the Laplacian kernel is defined as $\mathcal{K}(a, b) = \exp(-\|a - b\|_1 / \sigma)$, where $\sigma > 0$ is the bandwidth parameter, $\|\cdot\|_2$ is the usual Euclidean (ℓ_2 -)norm and $\|\cdot\|_1$ is the ℓ_1 -norm (Muandet et al., 2017, Table 3.1).

2 CI testing using conditional generators

In this section, we first describe a population-level conditional independence measure and present two of its sample versions, one of which benefits from a double robustness property against possible misspecification of the two marginal conditional distributions. We then propose a sample plug-in test statistic based on the superior version, utilizing the generative neural network technique to sample from the marginal conditional distributions along with data splitting and cross-fitting to remove certain data dependence. Following that, we provide a computational framework and a wild bootstrap method to calibrate the size of the proposed testing procedure. Some discussions on the connections between our proposed test statistic and existing CI tests in the literature are provided in [Appendix A of the online supplementary material](#).

2.1 Population-level conditional independence measure

Recall that $X \in \mathbb{R}^{d_X}$, $Y \in \mathbb{R}^{d_Y}$, and $Z \in \mathbb{R}^{d_Z}$ are three random vectors defined on the same probability space, and the goal is to test

$$H_0: X \perp\!\!\!\perp Y | Z \quad \text{vs.} \quad H_1: X \not\perp\!\!\!\perp Y | Z. \quad (1)$$

In order to motivate the construction of our population-level conditional independence measure, it is helpful to introduce two new random vectors $\tilde{X}$ and $\tilde{Y}$ which satisfy all of the properties in the following lemma (see [Appendix K.1 in the online supplementary material for proof](#)).

Lemma 1 Let $(X, Y, Z) \in \mathbb{R}^{d_X \times d_Y \times d_Z}$ be three random vectors with joint distribution P_{XYZ} . Then, there exist two random vectors $\tilde{X} \in \mathbb{R}^{d_X}$ and $\tilde{Y} \in \mathbb{R}^{d_Y}$ such that:

1. $(\tilde{X}, \tilde{Y})$ are conditionally independent of (X, Y) given Z ;
2. $\tilde{X}$ and $\tilde{Y}$ are conditionally independent given Z ;
3. $P_{\tilde{X}|Z=z} = P_{X|Z=z}$ and $P_{\tilde{Y}|Z=z} = P_{Y|Z=z}$ for a.e. $-P_Z$ $z \in \mathbb{R}^{d_Z}$.

Note that for any random vectors $\tilde{X}$ and $\tilde{Y}$ satisfying Property 3 in Lemma 1, we have $P_{\tilde{X}} = P_{XZ}$ and $P_{\tilde{Y}} = P_{YZ}$, and therefore according to Property 2 of the same lemma, testing problem (1) can be equivalently formulated into the following distribution equality testing problem

$$H_0: P_{XYZ} = P_{\tilde{X}\tilde{Y}Z} \quad \text{vs.} \quad H_1: P_{XYZ} \neq P_{\tilde{X}\tilde{Y}Z}. \quad (2)$$

It is then natural to propose a population-level measure to quantify the discrepancy between P_{XYZ} and $P_{\tilde{X}\tilde{Y}Z}$. In order to mitigate the curse of dimensionality, we adopt a kernel-based measure using

MMD, which compares the (kernel) mean embeddings of P_{XYZ} and $P_{\tilde{X}\tilde{Y}\tilde{Z}}$ in an RKHS. We call the resulting measure as an MMD-based CI measure, or MMDCI in short. More specifically, we let $\mathcal{K}_X: \mathbb{R}^{d_X} \times \mathbb{R}^{d_X}$, $\mathcal{K}_Y: \mathbb{R}^{d_Y} \times \mathbb{R}^{d_Y}$, and $\mathcal{K}_Z: \mathbb{R}^{d_Z} \times \mathbb{R}^{d_Z}$ denote three symmetric positive definite kernel functions which define three RKHSs $\mathbb{H}_X$, $\mathbb{H}_Y$, and $\mathbb{H}_Z$ over the space of X , Y , and Z , respectively. We also let $\mathcal{K}_0 = \mathcal{K}_X \otimes \mathcal{K}_Y \otimes \mathcal{K}_Z$ and $\mathbb{H}_0$ denote the corresponding (product) kernel and its induced RKHS over the product space $\mathbb{R}^{d_X} \times \mathbb{R}^{d_Y} \times \mathbb{R}^{d_Z}$.

We use the notation $\langle \cdot, \cdot \rangle_{\mathbb{H}}$ and $\| \cdot \|_{\mathbb{H}}$ to denote the associated inner product and the induced norm of a generic RKHS $\mathbb{H}$ with kernel $\mathcal{K}$. For a generic random variable W taking values in the domain of $\mathbb{H}$, its (kernel) mean embedding into $\mathbb{H}$ is denoted as $\mathbb{E}[\mathcal{K}(W, \cdot)]$, which is defined as the unique element in $\mathbb{H}$ such that $\mathbb{E}[f(W)] = \langle f, \mathbb{E}[\mathcal{K}(W, \cdot)] \rangle_{\mathbb{H}}$ for any $f \in \mathbb{H}$. With these notations, we can now define the aforementioned MMDCI as

$$\text{MMDCI}(P_{XYZ}) := \left\| \mathbb{E}[\mathcal{K}_0((X, Y, Z), \cdot)] - \mathbb{E}[\mathcal{K}_0((\tilde{X}, \tilde{Y}, \tilde{Z}), \cdot)] \right\|_{\mathbb{H}_0}^2 \quad (3)$$

$$\begin{aligned} &= \mathbb{E}[\mathcal{K}_0((X, Y, Z), (X', Y', Z'))] - 2\mathbb{E}[\mathcal{K}_0((X, Y, Z), (\tilde{X}', \tilde{Y}', Z'))] \\ &\quad + \mathbb{E}[\mathcal{K}_0((\tilde{X}, \tilde{Y}, \tilde{Z}), (\tilde{X}', \tilde{Y}', Z'))], \end{aligned} \quad (4)$$

where $(X', Y', \tilde{X}', \tilde{Y}', Z')$ denotes an independent copy of $(X, Y, \tilde{X}, \tilde{Y}, Z)$, and the second equality (4) follows by expanding the squared RKHS norm, using the definition of the mean embedding and applying the reproducing property of the RKHS $\mathbb{H}_0$. Note that this MMDCI measure only depends on the joint distribution P_{XYZ} , as $P_{\tilde{X}\tilde{Y}\tilde{Z}}$ is implicitly defined through P_{XYZ} using Lemma 1. In particular, we have the following Lemma 2 (which is proved in [Appendix K.1 of the online supplementary material](#)) relating $\text{MMDCI}(P_{XYZ})$ to H_0 in problem (1) or (2) if Assumption 1 holds.

Assumption 1 For each $\tilde{Q} \in \{\tilde{X}, X\}$, assume either one of the following assumptions holds:

- (a) $\mathbb{E}[\sqrt{\mathcal{K}_X(\tilde{Q}, \tilde{Q}) \mathcal{K}_Y(Y, Y) \mathcal{K}_Z(Z, Z)}] < \infty$. $\mathcal{K}_0$ is a characteristic kernel ([Fukumizu et al., 2007](#), Section 2.2), meaning that for any probability measures P and Q on $\mathbb{R}^{d_X+d_Y+d_Z}$, $\mathbb{E}_{(X,Y,Z) \sim P}[\mathcal{K}_0((X, Y, Z), \cdot)] = \mathbb{E}_{(X,Y,Z) \sim Q}[\mathcal{K}_0((X, Y, Z), \cdot)]$ implies $P = Q$.
- (b) $\mathbb{H}_X \otimes \mathbb{H}_Z$ is dense in $L_2(\mathbb{R}^{d_X+d_Z}, P_{XZ})$ and $\mathbb{H}_Y$ is dense in $L_2(\mathbb{R}^{d_Y}, P_Y)$.
- (c) $\mathbb{E}[\mathcal{K}_X(\tilde{Q}, \tilde{Q}) \mathcal{K}_Y(Y, Y)] < \infty$, $\mathbb{H}_Z$ is dense in $L_2(\mathbb{R}^{d_Z}, P_Z)$ and $(\mathcal{K}_X, \mathcal{K}_Y)$ are continuous characteristic kernels.

Lemma 2 Let $(X, Y, Z) \in \mathbb{R}^{d_X \times d_Y \times d_Z}$ be three random vectors with joint distribution P_{XYZ} . Under Assumption 1, X and Y are conditionally independent given Z if and only if $\text{MMDCI}(P_{XYZ}) = 0$.

Remark 1 Part (a) of Assumption 1 comes from the definition of MMDCI in equation (3). Part (b) of Assumption 1 comes from the CI characterization in [Daudin \(1980\)](#) (see [Equation \(A-5\) in Appendix A.3, online supplementary material](#)), and it is implied by $\mathcal{K}_X \otimes \mathcal{K}_Z$ and $\mathcal{K}_Y$ being L_2 - or c_0 -universal kernels ([Sriperumbudur et al., 2011](#)), and it can be alternatively stated as: $\mathbb{H}_Y \otimes \mathbb{H}_Z$ is dense in $L_2(\mathbb{R}^{d_Y+d_Z}, P_{YZ})$ and $\mathbb{H}_X$ is dense in $L_2(\mathbb{R}^{d_X}, P_X)$. According to [Sriperumbudur et al. \(2011\)](#) and Theorems 4 and 5 in [Szabó and Sriperumbudur \(2018\)](#), Assumption 1 holds if $(\mathcal{K}_X, \mathcal{K}_Y, \mathcal{K}_Z)$ are Gaussian or

Laplacian kernels. Part (a) is in general more restrictive than part (c) because it implies $(\mathcal{K}_X, \mathcal{K}_Y, \mathcal{K}_Z)$ are all characteristic kernels. If $(\mathcal{K}_X, \mathcal{K}_Y, \mathcal{K}_Z)$ are bounded, continuous and translation invariant kernels, both parts (a) and (c) are equivalent to $(\mathcal{K}_X, \mathcal{K}_Y, \mathcal{K}_Z)$ being characteristic kernels (Szabó & Sriperumbudur, 2018, Theorem 4).

In practice, in order to facilitate the construction of a sample version of $\text{MMDCI}(P_{XYZ})$ defined in (4), it is helpful to eliminate the tilde variables $(\tilde{X}, \tilde{Y}, \tilde{X}', \tilde{Y}')$ by taking conditional expectations given (Z, Z') and utilizing the iterative expectation formula and their mutual conditional independence. Specifically, we let $g_X(z) = \mathbb{E}[\mathcal{K}_X(X, \cdot) | Z = z] \in \mathbb{H}_X$ and $g_Y(z) = \mathbb{E}[\mathcal{K}_Y(Y, \cdot) | Z = z] \in \mathbb{H}_Y$ for any $z \in \mathbb{R}^{d_Z}$ denote the *conditional (kernel) mean embedding* of X and Y given Z into their respective RKHS $\mathbb{H}_X$ and $\mathbb{H}_Y$. The simplest way to eliminate the tilde variables is to use the factorization property $\mathcal{K}_0 = \mathcal{K}_X \otimes \mathcal{K}_Y \otimes \mathcal{K}_Z$ along with the iterative expectation formula to (4) to get (see Appendix K.1 in the online supplementary material for a detailed derivation)

$$\begin{aligned} \text{MMDCI}(P_{XYZ}) = \mathbb{E} \left\{ \left[\mathcal{K}_X(X, X') \cdot \mathcal{K}_Y(Y, Y') - \langle g_X(Z'), \mathcal{K}_X(X, \cdot) \rangle_{\mathbb{H}_X} \cdot \langle g_Y(Z'), \mathcal{K}_Y(Y, \cdot) \rangle_{\mathbb{H}_Y} \right. \right. \\ \left. - \langle g_X(Z), \mathcal{K}_X(X', \cdot) \rangle_{\mathbb{H}_X} \cdot \langle g_Y(Z), \mathcal{K}_Y(Y', \cdot) \rangle_{\mathbb{H}_Y} \right. \\ \left. + \langle g_X(Z), g_X(Z') \rangle_{\mathbb{H}_X} \cdot \langle g_Y(Z), g_Y(Z') \rangle_{\mathbb{H}_Y} \right] \cdot \mathcal{K}_Z(Z, Z') \right\}, \end{aligned} \quad (5)$$

where we deliberately express the quantity inside $\text{MMDCI}(P_{XYZ})$ in a way that is symmetric with respect to (X, Y, Z) and its independent copy (X', Y', Z') . With the knowledge of the two conditional distributions $P_{X|Z}$ and $P_{Y|Z}$, or their mean embeddings g_X and g_Y , a sample version of $\text{MMDCI}(P_{XYZ})$ based on equation 5, due to the symmetry, can then be constructed as a degenerate U-statistic of order 2. Regarding the two conditional distributions, we propose using a GNN framework to approximate their conditional expectations via Monte Carlo sampling, and then plugging-in these conditional expectation estimates into the degenerate U-statistic to construct our sample version of the MMDCI test statistic; see Section 2.3 for details.

2.2 A doubly robust population-level measure characterization

In order to motivate our construction of a sample version of the MMDCI test statistic that enjoys a desirable double robustness property (see Definition 1), it is instructive to provide an alternative and equivalent characterization of $\text{MMDCI}(P_{XYZ})$ as summarized in the following lemma, which is proved in Appendix K.2 in the online supplementary material. This alternative characterization can be obtained from equation 4 by first conditioning on (Z, Z') and then utilizing the factorization property $\mathcal{K}_0 = \mathcal{K}_X \otimes \mathcal{K}_Y \otimes \mathcal{K}_Z$ and the reproducing property of the RKHSs.

Lemma 3 Let $(X, Y, Z) \in \mathbb{R}^{d_X \times d_Y \times d_Z}$ be three random vectors with joint distribution P_{XYZ} and (X', Y', Z') be its independent copy. Then the $\text{MMDCI}(P_{XYZ})$ defined in equation 4 can be equivalently written as

$$\begin{aligned} \text{MMDCI}(P_{XYZ}) = \mathbb{E} \left[U(X, X') V(Y, Y') \mathcal{K}_Z(Z, Z') \right], \quad \text{with} \\ U(X, X') = \mathcal{K}_X(X, X') - \langle g_X(Z), \mathcal{K}_X(X', \cdot) \rangle_{\mathbb{H}_X} - \langle g_X(Z'), \mathcal{K}_X(X, \cdot) \rangle_{\mathbb{H}_X} \\ + \langle g_X(Z), g_X(Z') \rangle_{\mathbb{H}_X} \\ V(Y, Y') = \mathcal{K}_Y(Y, Y') - \langle g_Y(Z), \mathcal{K}_Y(Y', \cdot) \rangle_{\mathbb{H}_Y} - \langle g_Y(Z'), \mathcal{K}_Y(Y, \cdot) \rangle_{\mathbb{H}_Y} \\ + \langle g_Y(Z), g_Y(Z') \rangle_{\mathbb{H}_Y}. \end{aligned} \quad (6)$$

Recall that $g_X(z) = \mathbb{E}[\mathcal{K}_X(X, \cdot) | Z = z] \in \mathbb{H}_X$ and $g_Y(z) = \mathbb{E}[\mathcal{K}_Y(Y, \cdot) | Z = z] \in \mathbb{H}_Y$ for any $z \in \mathbb{R}^{d_Z}$.

This alternative characterization of $\text{MMDCI}(P_{XYZ})$ takes a multiplicative form and eliminates the need of introducing $(\tilde{X}, \tilde{Y})$ (and its independent copy). Moreover, its constituting multiplicative factors $U(X, X')$ and $V(Y, Y')$ are both zero mean random variables and symmetric in (X, X') and (Y, Y') , respectively. In particular, under H_0 , $U(X, X')$ and $V(Y, Y')$ also become conditionally independent given (Z, Z') , implying $\text{MMDCI}(P_{XYZ})$ to be zero under H_0 . Due to the symmetry, one can also construct a sample version of the MMDCI test statistic based on [equation 6](#) as a degenerate U-statistic of order 2; see the comments after [equation 5](#) and [Section 2.3](#) for further details.

These desirable properties together would lead to a *double robustness* property of this population-level measure $\text{MMDCI}(P_{XYZ})$ expressed from this alternative characterization [\(6\)](#) against potential misspecification of the two (marginal) conditional distributions $P_{X|Z}$ and $P_{Y|Z}$ in estimating g_X and g_Y . More generally, we formally define such a population-level double robustness property for any population-level measure that is defined as functions of (X, Y, Z, X', Y', Z') and conditional expectations in the form of $\mathbb{E}[u(X)|Z]$ and/or $\mathbb{E}[v(Y)|Z]$ for certain mappings $u: \mathbb{R}^{d_Z} \rightarrow \mathbb{U}$ and $v: \mathbb{R}^{d_Z} \rightarrow \mathbb{V}$ with $(\mathbb{U}, \mathbb{V})$ denoting two generic Hilbert spaces as ranges of (u, v) as follows.

Definition 1 Consider a generic population-level measure taking the form as $D(P_{XYZ}) = \mathbb{E}[F(X, Y, Z, X', Y', Z', \mathbb{E}[u(X)|Z], \mathbb{E}[u(X)|Z'], \mathbb{E}[v(Y)|Z], \mathbb{E}[v(Y)|Z'])]$ for some known mappings $u: \mathbb{R}^{d_Z} \rightarrow \mathbb{U}$, $v: \mathbb{R}^{d_Z} \rightarrow \mathbb{V}$, and $F: \mathbb{R}^{2d_X+2d_Y+2d_Z} \times \mathbb{U}^2 \times \mathbb{V}^2 \rightarrow \mathbb{R}$. We say that $D(P_{XYZ})$ is doubly robust against misspecifications of the two conditional expectations $g_u(\cdot) = \mathbb{E}[u(X)|Z = \cdot]$ and $g_v(\cdot) = \mathbb{E}[v(Y)|Z = \cdot]$ if under H_0 of testing problem [\(1\)](#), $D(P_{XYZ})$ equals to zero as long as at least one of these two conditional expectations is correctly specified; that is, $\mathbb{E}[F(X, Y, Z, X', Y', Z', g'_u(Z), g'_u(Z'), g'_v(Z), g'_v(Z')))] = 0$ if either $g'_u = g_u$ or $g'_v = g_v$.

Double robustness is in general a desirable property in many statistical problems where certain estimators or testing procedures maintain their consistency or validity when either of two involved models is correctly specified, but not necessarily both. This property is particularly useful in causal inference and missing data analysis (see, e.g. [Bang & Robins, 2005](#); [Funk et al., 2011](#)) which involves a treatment model and an outcome model. In our definition of double robustness for CI testing, the two involved models are the two approximation families for the two conditional distributions $P_{X|Z}$ and $P_{Y|Z}$ (or more precisely, the conditional expectations g_u and g_v for certain transformations u and v). A theoretical consequence of this population-level double robustness property from [Definition 1](#) on the asymptotic analysis of a sample version of the test statistic is that, the test statistic will retain all desirable properties of the oracle test statistic, defined as the one using the true marginal conditional distributions, as long as the product of the two statistical errors for estimating $P_{X|Z}$ and $P_{Y|Z}$ (or the relevant conditional expectations g_u and g_v) decays to zero faster than the parametric rate; see [Section 3](#) for details. This double robustness property provides a safeguard against model misspecification and allows more flexibility and tolerance in estimating the two marginal conditional distributions, making our testing procedure less sensitive to the estimation errors, which typically suffer from a slower non-parametric decay rate.

In our current context, it is straightforward to verify that $\text{MMDCI}(P_{XYZ})$ in [Lemma 3](#) takes the form as in [Definition 1](#); moreover, it is doubly robust against misspecifications of the two conditional mean embeddings $g_X(z) = \mathbb{E}[\mathcal{K}_X(X, \cdot) | Z = z] \in \mathbb{H}_X := \mathbb{U}$ and $g_Y(z) = \mathbb{E}[\mathcal{K}_Y(Y, \cdot) | Z = z] \in \mathbb{H}_Y := \mathbb{V}$ since $U(X, X')$ and $V(Y, Y')$ are both zero mean random variables that are conditionally independent given (Z, Z') under H_0 . In comparison, under H_0 the $\text{MMDCI}(P_{XYZ})$ in its earlier representation [\(5\)](#) can also be written in the form as in [Definition 1](#); however, it is not doubly robust against misspecifications of the two mean embeddings. To see this, consider replacing $g_X(z)$ in representation [\(5\)](#) with $\hat{g}_X(z) = \mathbb{E}[\mathcal{K}_X(\hat{X}, \cdot) | Z = z]$, where $\hat{X}$ is any random vector satisfying $\hat{X} \perp\!\!\!\perp Y | Z$ and $P_{\hat{X}|Z} \neq P_{X|Z}$; then according to [Lemma 7 in Appendix K.3, online supplementary material](#), $\text{MMDCI}(P_{XYZ})$ after the replacement becomes $\|\mathbb{E}[\mathcal{K}_0((X, Y, Z), \cdot)] - \mathbb{E}[\mathcal{K}_0((\hat{X}, Y, Z), \cdot)]\|_{\mathbb{H}_0}^2$, which is the squared MMD between P_{XYZ} and $P_{\hat{X}|Z}$ and is strictly positive when $\mathcal{K}_0$ is a

characteristic kernel. See also Section 2.3 for discussions on the consequences in the sample testing statistic accuracy, and Section 3 for theoretical implications. The major reason for the same population-level measure having or lacking the double robustness property under different representations is attributed to how $(\tilde{X}, \tilde{Y}, \tilde{X}', \tilde{Y}')$ are eliminated by taking conditional expectations (on which terms) given (Z, Z') .

Note that our alternative characterization (6) generalizes the one derived in Cai et al. (2022) in the special case when $X, Y, Z \sim \text{Uniform}[0, 1]$, $X \perp\!\!\!\perp Z$, $Y \perp\!\!\!\perp Z$ and $(\mathcal{K}_X, \mathcal{K}_Y, \mathcal{K}_Z)$ are the Laplacian kernel. This characterization (6) is closely related to the conditional independence measure proposed in Daudin (1980), which is also later used in K. Zhang et al. (2011), Pogodin et al. (2022), and Pogodin et al. (2024), to construct kernel-based CI test statistics. See Appendix A.3, online supplementary material for more details and comparisons.

2.3 Sample conditional independence test statistic

In this subsection, we provide sample versions arising from the two equivalent characterizations (5) and (6) for $\text{MMDCI}(P_{XYZ})$, and discuss the impact of, or the lack of, the double robustness property defined in Definition 1 on their respective statistical accuracy. In practical calculations, it is easier to utilize the following identities, arising from the reproducing property of the RKHS,

$$\begin{aligned} \langle g_X(Z), \mathcal{K}_X(X', \cdot) \rangle_{\mathbb{H}_X} &= \mathbb{E}_X[\mathcal{K}_X(X, X') | Z] \quad \text{and} \\ \langle g_X(Z), g_X(Z') \rangle_{\mathbb{H}_X} &= \mathbb{E}_{X, X'}[\mathcal{K}_X(X, X') | Z, Z']. \end{aligned} \quad (7)$$

to further simplify terms involving the conditional mean embeddings g_X and g_Y in these characterizations. More concretely, we first consider the case when $P_{X|Z}$ and $P_{Y|Z}$ (or their corresponding conditional mean embeddings) are known, and construct their sample versions as degenerate U-statistics of order 2. We call these test statistics as *oracle test statistics* since later we will consider the general and practical case, where $P_{X|Z}$ and $P_{Y|Z}$ are unknown and estimated from the data. Let $\{(X_i, Y_i, Z_i)\}_{i=1}^n$ denote i.i.d. copies of (X, Y, Z) from P_{XYZ} .

2.3.1 Oracle test statistic based on characterization (5)

It is easy to see that characterization (5) of $\text{MMDCI}(P_{XYZ})$ leads to the following oracle test statistic,

$$\begin{aligned} T_0^* &= \frac{1}{n(n-1)} \sum_{k \neq \ell} \mathcal{K}_Z(Z_k, Z_\ell) \cdot \{ \mathcal{K}_X(X_k, X_\ell) \cdot \mathcal{K}_Y(Y_k, Y_\ell) - \mathbb{E}_X[\mathcal{K}_X(X, X_\ell) | Z = Z_k] \\ &\quad \cdot \mathbb{E}_Y[\mathcal{K}_Y(Y, Y_\ell) | Z = Z_k] - \mathbb{E}_X[\mathcal{K}_X(X, X_k) | Z = Z_\ell] \cdot \mathbb{E}_Y[\mathcal{K}_Y(Y, Y_k) | Z = Z_\ell] \\ &\quad + \mathbb{E}_{X, X'}[\mathcal{K}_X(X, X') | Z = Z_k, Z' = Z_\ell] \cdot \mathbb{E}_{Y, Y'}[\mathcal{K}_Y(Y, Y') | Z = Z_k, Z' = Z_\ell] \}. \end{aligned} \quad (8)$$

2.3.2 Oracle test statistic based on characterization (6)

Basing on identity (7), we can define the following statistic from characterization (6),

$$\begin{aligned} T^* &= \frac{1}{n(n-1)} \sum_{k \neq \ell} U^*(X_k, X_\ell) V^*(Y_k, Y_\ell) \mathcal{K}_Z(Z_k, Z_\ell), \quad \text{with} \\ U^*(X_k, X_\ell) &= \mathcal{K}_X(X_k, X_\ell) - \mathbb{E}_X[\mathcal{K}_X(X, X_k) | Z = Z_\ell] - \mathbb{E}_X[\mathcal{K}_X(X, X_\ell) | Z = Z_k] \\ &\quad + \mathbb{E}_{X, X'}[\mathcal{K}_X(X, X') | Z = Z_k, Z' = Z_\ell], \\ V^*(Y_k, Y_\ell) &= \mathcal{K}_Y(Y_k, Y_\ell) - \mathbb{E}_Y[\mathcal{K}_Y(Y, Y_k) | Z = Z_\ell] - \mathbb{E}_Y[\mathcal{K}_Y(Y, Y_\ell) | Z = Z_k] \\ &\quad + \mathbb{E}_{Y, Y'}[\mathcal{K}_Y(Y, Y') | Z = Z_k, Z' = Z_\ell]. \end{aligned} \quad (9)$$

Under H_0 and suitable assumptions, both nT_0^* and nT^* converge in distribution (after rescaling by the same size) to the typical limiting null distribution of a degenerate U-statistic, which is a weighted sum of i.i.d. centred χ_1^2 random variables. However, they have different limiting distributions; see Appendix D, online supplementary material for an empirical comparison between their limiting null distributions and powers of tests based on T_0^* and T^* . Specifically, we observe

that the test based on T^* tends to produce a higher (size-adjusted) power than T_0^* under conditional dependence since T^* tends to exhibit a smaller asymptotic variance.

2.3.3 Conditional mean embedding estimation via generative neural networks

In practice, since $P_{X|Z}$ and $P_{Y|Z}$ are rarely known, we need to find estimators $\hat{g}_X(z)$ and $\hat{g}_Y(z)$ for the two conditional mean embeddings $g_X(z) \in \mathbb{H}_X$ and $g_Y(z) \in \mathbb{H}_Y$ for each $z \in \mathbb{R}^{d_Z}$. In the literature, [Pogodin et al. \(2024\)](#) propose to estimate them via RKHS-valued kernel ridge regression (see [Appendix A.3, online supplementary material](#) for details), which may suffer from slow convergence rate due to the curse of dimensionality on d_Z (see [Appendix C, online supplementary material](#) for an empirical comparison). In this work, in order to mitigate this issue, we adopt a generative neural network (GNN) framework to instead train two conditional generators $\hat{\mathbb{G}}_X: \mathbb{R}^m \times \mathbb{R}^{d_Z} \rightarrow \mathbb{R}^{d_X}$ and $\hat{\mathbb{G}}_Y: \mathbb{R}^m \times \mathbb{R}^{d_Z} \rightarrow \mathbb{R}^{d_Y}$ for approximately sampling from the respective conditional distributions $P_{X|Z}$ and $P_{Y|Z}$. Concretely, if one sample a latent variable η from some simple distribution (such as the standard normal) over the latent space $\mathbb{R}^m$ for some integer $m \geq 1$, then the conditional distributions of $\hat{X} = \hat{\mathbb{G}}_X(\eta, Z)$ and $\hat{Y} = \hat{\mathbb{G}}_Y(\eta, Z)$ given Z (denoted as $\hat{P}_{X|Z}$ and $\hat{P}_{Y|Z}$) are good approximations of $P_{X|Z}$ and $P_{Y|Z}$. In fact, according to the noise-outsourcing lemma (see Theorem 6.10 of [Kallenberg, 2002](#), also Lemma 2.1 of [X. Zhou et al., 2022](#)), for any integer $m \geq 1$, there exist measurable functions $\mathbb{G}_X$ and $\mathbb{G}_Y$ such that for any $\eta \sim N(0, I_m)$ that is independent of Z , we have $\mathbb{G}_X(\eta, Z) | Z \sim P_{X|Z}$ and $\mathbb{G}_Y(\eta, Z) | Z \sim P_{Y|Z}$. To further estimate $g_X(z)$ for any $z \in \mathbb{R}^{d_Z}$, one can first generate M i.i.d. samples of $\{\eta_i\}_{i=1}^M$ from $N(0, I_m)$, and then estimate $g_X(z)$ by the sample average $\hat{g}_X(z) := M^{-1} \sum_{i=1}^M \mathcal{K}_X(\mathbb{G}_X(\eta_i, z), \cdot) \in \mathbb{H}_X$. Likewise, we can define estimator $\hat{g}_Y(z) \in \mathbb{H}_Y$ for $g_Y(z)$ in a completely analogous manner.

There are many GNN methods available for approximating conditional distributions. We adopt the GMMN framework due to its competitive performance and a close connection (minimizing an MMD) with our proposed test (see Remark 3). Concretely, under our notations, the conditional generator $\hat{\mathbb{G}}$ in GMMN for approximating $P_{X|Z}$ ($P_{Y|Z}$ is treated in an analogous way) is obtained by minimizing the sample version of a squared MMD between P_{XZ} and the induced joint distribution $\hat{P}_{XZ}$ from the estimated $\hat{X} = \hat{\mathbb{G}}_X(\eta, Z)$ based on a generic set of training data $\{(X_i, Z_i)\}_{i=1}^{n_T}$ with training sample size n_T and Mn_T latent variables $\{\eta_i^m : i = 1, \dots, n_T, m = 1, \dots, M\}$:

$$\begin{aligned} \hat{\mathbb{G}}_X &= \arg \min_{\mathbb{G}_X \in \mathcal{G}_X} \frac{1}{n_T(n_T-1)} \sum_{\substack{k, \ell \in [n_T] \\ k \neq \ell}} \hat{U}(X_k, X_\ell) \cdot \mathcal{K}_Z(Z_k, Z_\ell), \\ \text{with } \hat{U}(X_k, X_\ell) &= \mathcal{K}_X(X_k, X_\ell) - \frac{1}{M} \sum_{m=1}^M \mathcal{K}_X(X_k, \mathbb{G}_X(\eta_\ell^m, Z_\ell)) \\ &\quad - \frac{1}{M} \sum_{m=1}^M \mathcal{K}_X(X_\ell, \mathbb{G}_X(\eta_k^m, Z_k)) \\ &\quad + \frac{1}{M^2} \sum_{m_1, m_2=1}^M \mathcal{K}_X(\mathbb{G}_X(\eta_k^{m_1}, Z_k), \mathbb{G}_X(\eta_\ell^{m_2}, Z_\ell)), \end{aligned} \tag{10}$$

where $\mathcal{G}_X$ is an approximation family, such as (deep) neural networks, for the conditional generators. If $\mathcal{K}_X \otimes \mathcal{K}_Z$ is a bounded characteristic kernel and there exist $\mathbb{G}_X^* \in \mathcal{G}_X$ such that $(\mathbb{G}_X^*(\eta, Z), Z) \stackrel{d}{=} (X, Z)$, the generalization bound of GMMN ([Dziugaite et al., 2015](#), Theorem 1) guarantees that the MMD with kernel $\mathcal{K}_X \otimes \mathcal{K}_Z$ between the distributions of $(\hat{\mathbb{G}}_X(\eta, Z), Z)$ and (X, Z) converges to zero at the rate n_T^{-r} ; here, constant $r > 0$ typically depends on smoothness properties of $\mathbb{G}_X^*$, intrinsic dimensionality of X and/or Z , choices of the kernels and some complexity measure of $\mathcal{G}_X$, and can be as large as $1/2$ (corresponding to a parametric convergence rate). Note that in the original GMMN framework in [Dziugaite et al. \(2015\)](#), the Monte Carlo sample size M is fixed at one, while our numerical results suggest that choosing a larger M (e.g. in the order of n_T) can dramatically improve the empirical performance and stability of the training process.

Specific designs of the neural network architectures for $\mathcal{G}_X$ in our GMMN implementation and choices of other tuning parameter values in the training process are provided in [Appendix G, online supplementary material](#).

2.3.4 Comparison of two oracle test statistics

In this part, section we compare the two oracle test statistics (8) and (9) by examining the impact of, or the lack of, the double robustness property, as defined in Definition 1, on their respective statistical accuracy after plugging-in estimated conditional mean embeddings.

To formally study the impact of plugging-in inaccurate conditional mean embeddings, we consider two generic error metrics $D_X(g_X, \hat{g}_X)$ and $D_Y(g_Y, \hat{g}_Y)$ to quantify the estimation errors of the estimators $\hat{g}_X$ and $\hat{g}_Y$ for the conditional mean embeddings. For example, when all employed kernels are bounded, we can choose $D_X(g_X, g'_X) = \{\mathbb{E}[\|g_X(Z) - g'_X(Z)\|_{\mathbb{H}_X}^2]\}^{1/2}$ (recall that for each $z \in \mathbb{R}^{d_z}$, we have $g_X(z) \in \mathbb{H}_X$) and $D_Y(g_Y, g'_Y) = \{\mathbb{E}[\|g_Y(Z) - g'_Y(Z)\|_{\mathbb{H}_Y}^2]\}^{1/2}$ in our theoretical analysis in Section 3. Recall that we used n_T to denote the GNN training sample size. Without loss of generality, we may assume $D_X(g_X, \hat{g}_X) = O(n_T^{-\alpha_1})$ and $D_Y(g_Y, \hat{g}_Y) = O(n_T^{-\alpha_2})$ for some nonnegative constants (α_1, α_2) . Here, (α_1, α_2) characterizes the respective estimation error decay exponent and depends on the GNN training algorithm and properties of the truth g_X and g_Y . Despite the fact that GNN training, such as by the aforementioned GMMN method, is capable of approximating complex distributions, the estimation error of the resulting estimators usually decay slower than the parametric $n_T^{-1/2}$ rate due to its non-parametric nature.

Now suppose that we have two folds of mutually independent samples, with n_T training samples for estimating (g_X, g_Y) and n fitting samples for constructing the two oracle test statistic T_0^* and T^* as defined in (8) and (9). Let $\hat{T}_0 = T_0(\hat{g}_X, \hat{g}_Y)$ and $\hat{T} = T(\hat{g}_X, \hat{g}_Y)$ denote the respective plug-in test statistic from the oracle ones T_0^* and T^* obtained by replacing g_X and g_Y (or the corresponding conditional expectations given Z arising from identity (7)) therein with estimators $\hat{g}_X$ and $\hat{g}_Y$. Using this notation, we can also write $T_0^* = T_0(g_X, g_Y)$ and $T^* = T(g_X, g_Y)$. Note that we have $\mathbb{E}[T_0(g_X, g_Y)] = \mathbb{E}[T(g_X, g_Y)] = 0$ under the null H_0 due to their constructions. Therefore, without any additional structures, a typical perturbation analysis gives that under H_0 ,

$$\begin{aligned} \hat{T}_0 &= T_0^* + \underbrace{\{T_0(\hat{g}_X, g_Y) - T_0(g_X, g_Y)\}}_{=I_1^*} + \underbrace{\{T_0(g_X, \hat{g}_Y) - T_0(g_X, g_Y)\}}_{=I_2^*} \\ &\quad + \underbrace{\{T_0(\hat{g}_X, \hat{g}_Y) - T_0(\hat{g}_X, g_Y) - T_0(g_X, \hat{g}_Y) + T_0(g_X, g_Y)\}}_{=I_3^*} \\ &= T_0^* + O_p(n_T^{-\alpha_1} n^{-1/2} + n_T^{-2\alpha_1}) + O_p(n_T^{-\alpha_2} n^{-1/2} + n_T^{-2\alpha_2}) + O_p(n_T^{-\alpha_1 - \alpha_2}). \end{aligned}$$

Here, since the MMDCI(P_{XYZ}) in characterization (5) is not doubly robust, expectations ($\mathbb{E}[I_1^*]$, $\mathbb{E}[I_2^*]$, $\mathbb{E}[I_3^*]$) of the three terms (I_1^* , I_2^* , I_3^*) are generally not zero. As a result, as shown in [Lemma 7 in Appendix K.3, online supplementary material](#), I_1^* resembles a non-degenerate U-statistic (with nonzero mean) and satisfies $I_1^* = O_p(\sqrt{\text{Var}(I_1^*)} + \mathbb{E}[I_1^*]) = O_p(n_T^{-\alpha_1} n^{-1/2} + n_T^{-2\alpha_1})$ (similar results hold for I_2^*). For the ‘higher-order interaction term’ I_3^* , its decay rate satisfies $I_3^* = O_p(n_T^{-\alpha_1 - \alpha_2})$ and depends on both estimation errors. This means that in order for the plug-in error $\hat{T}_0 - T_0^*$ to be asymptotically negligible for approximating T_0^* , or $\hat{T}_0 - T_0^* = o_p(T_0^*)$ as $(n_T, n) \rightarrow \infty$, so that $\hat{T}_0$ can retain all desirable properties of the oracle test statistic T_0^* , we need $n_T^{-\alpha_1} n^{-1/2} + n_T^{-2\alpha_1} + n_T^{-\alpha_2} n^{-1/2} + n_T^{-2\alpha_2} + n_T^{-\alpha_1 - \alpha_2} = o(n^{-1})$, or $\min(\alpha_1, \alpha_2) > 1/2$ if n_T scales as the same order as n . In [Appendix B, online supplementary material](#), we empirically show that the plug-in test statistic $\hat{T}_0$ based on T_0^* indeed has non-diminishing size distortion, even when both estimators $(\hat{g}_X, \hat{g}_Y)$ attain a fast $n^{-1/2}$ rate of convergence when $n_T = n$ in a parametric setting. This implies that our previous perturbation analysis is tight and naively plugging in these estimators into a simple sample version of the population-level CI measure can lead to non-diminishing size distortion in testing H_0 .

Fortunately, our improved oracle statistic T^* as defined in (9) admits a better perturbation bound, because of its population-level double robustness property as defined in Definition 1. More specifically, we can follow the same decomposition to obtain

$$\begin{aligned}\widehat{T} &= T^* + \underbrace{\{T(\widehat{g}_X, g_Y) - T(g_X, g_Y)\}}_{=I_1^\dagger} + \underbrace{\{T(g_X, \widehat{g}_Y) - T(g_X, g_Y)\}}_{=I_2^\dagger} \\ &\quad + \underbrace{\{T(\widehat{g}_X, \widehat{g}_Y) - T(\widehat{g}_X, g_Y) - T(g_X, \widehat{g}_Y) + T(g_X, g_Y)\}}_{=I_3^\dagger} \\ &= T^* + O_p(n_T^{-\alpha_1} n^{-1}) + O_p(n_T^{-\alpha_2} n^{-1}) + O_p(n_T^{-\alpha_1-\alpha_2} n^{-1/2} + n_T^{-2(\alpha_1+\alpha_2)}),\end{aligned}$$

where now the decay rates of the two terms $I_1^\dagger$ and $I_2^\dagger$ have an extra factor of $n^{-1/2}$ compared to those in the earlier decomposition of $\widehat{T}_0$, due to the double robustness property $\mathbb{E}[I_1^\dagger] = \mathbb{E}[I_2^\dagger] = 0$. Specifically, from Lemma 6 in Appendix J, online supplementary material, $I_1^\dagger$ now resembles a degenerate U-statistic (with zero mean) and satisfies $I_1^\dagger = O_p(\sqrt{\text{Var}(I_1^\dagger)}) = O_p(n_T^{-\alpha_1} n^{-1})$ (similar results hold for $I_2^\dagger$). Additionally, although term $I_3^\dagger$ does not have a zero mean, it is a higher-order remainder term and also has a much faster decay rate than I_3^* , that is, $I_3^\dagger = O_p(n_T^{-\alpha_1-\alpha_2} n^{-1/2} + n_T^{-2(\alpha_1+\alpha_2)})$. This analysis implies that the plug-in test statistic based on T^* will have correct asymptotic size and attain the same asymptotic limit as T^* as long as the product of the two estimation errors decays faster than $n^{-1/2}$, or $\alpha_1 + \alpha_2 > 1/2$ if n_T scales as the same order as n . This requirement on the conditional distribution estimation accuracy is much weaker than the $\min(\alpha_1, \alpha_2) > 1/2$ requirement for the previous oracle statistic T_0^* , and allows both conditional mean embedding estimators $\widehat{g}_X$ and $\widehat{g}_Y$ to have non-parametric convergence rates slower than $n^{-1/2}$. For this reason, in the rest of this article, we will exclusively focus on the improved oracle statistic T^* and its plug-in version. See Appendix A.2, online supplementary material for a discussion on existing doubly robust CI tests.

2.4 Computation and size calibration via wild bootstrap

The analysis in Section 2.3.4 requires the training samples for estimating the conditional mean embeddings to be independent of the testing samples for constructing the sample test statistic. This motivates us to adopt a sample splitting and cross-fitting framework to remove dependence on conditional generator estimation and improve testing size accuracy. In this framework, we divide the entire dataset into J folds, using $(J-1)$ folds to train conditional generators and the remaining fold to construct a plug-in test statistic, and finally averaging the statistics over all J folds; see Section 4.1 for an empirical study of the impact of fold number J on the performance of the test statistic.

To be specific, let (η, κ) be two independent random variables from the latent variable distribution $N(0, I_m)$ on $\mathbb{R}^m$ that are also independent of (X, Y, Z) . Recall that we used $\mathbb{G}_X^*$ and $\mathbb{G}_Y^*$ to denote true conditional generators (see Section 2.3.3) that can generate samples from the two conditional distributions $P_{X|Z}$ and $P_{Y|Z}$. It is then easy to verify that $\widetilde{X} = \mathbb{G}_X^*(\eta, Z)$ and $\widetilde{Y} = \mathbb{G}_Y^*(\kappa, Z)$ serve as a valid pair of random vectors satisfying all properties in Lemma 1. Let $\mathcal{D} := \{(X_i, Y_i, Z_i)\}_{i=1}^n$ denote the dataset with sample size n , and $\{(\eta_i^m, \kappa_i^m) : i \in [n], m \in [M]\}$ be nM i.i.d. copies of (η, κ) such that $\{(X_i, Y_i, Z_i)\}_{i=1}^n$, $\{(\eta_i^1, \kappa_i^1)\}_{i=1}^n, \dots, \{(\eta_i^M, \kappa_i^M)\}_{i=1}^n$ are mutually independent. We divide $\{1, 2, \dots, n\}$ into J equal folds $\mathcal{J}^{(1)}, \dots, \mathcal{J}^{(J)}$ and let $\mathcal{J}^{(-j)} = \{1, 2, \dots, n\} \setminus \mathcal{J}^{(j)}$. For each $j \in [J]$, we use $\widehat{\mathbb{G}}_X^{(j)}$ and $\widehat{\mathbb{G}}_Y^{(j)}$ to denote the conditional generators trained through (10) using data from $\mathcal{D}_{-j} := \{(X_i, Y_i, Z_i)\}_{i \in \mathcal{J}^{(-j)}}$. For each $j \in [J]$, $i \in \mathcal{J}^{(j)}$ and $m \in [M]$, let $\widehat{X}_i^{(m)} = \widehat{\mathbb{G}}_X^{(j)}(\eta_i^m, Z_i)$ and $\widehat{Y}_i^{(m)} = \widehat{\mathbb{G}}_Y^{(j)}(\kappa_i^m, Z_i)$ if $i \in \mathcal{J}^{(j)}$. We can then define our final test statistic (based on the doubly robust oracle test statistic) as

$$\widehat{T}_J = \frac{1}{J} \sum_{j=1}^J \left\{ \frac{1}{\frac{n}{J} \left(\frac{n}{J} - 1 \right)} \sum_{\substack{k \neq \ell \\ k, \ell \in \mathcal{J}^{(j)}}} \widehat{U}(X_k, X_\ell) \widehat{V}(Y_k, Y_\ell) \mathcal{K}_Z(Z_k, Z_\ell) \right\}, \quad (11)$$

where

$$\begin{aligned}\widehat{U}(X_k, X_\ell) &= \mathcal{K}_X(X_k, X_\ell) - \frac{1}{M} \sum_{m=1}^M \mathcal{K}_X(X_k, \widehat{X}_\ell^{(m)}) - \frac{1}{M} \sum_{m=1}^M \mathcal{K}_X(X_\ell, \widehat{X}_k^{(m)}) \\ &\quad + \frac{1}{M^2} \sum_{m_1, m_2=1}^M \mathcal{K}_X(\widehat{X}_k^{(m_1)}, \widehat{X}_\ell^{(m_2)}), \\ \widehat{V}(Y_k, Y_\ell) &= \mathcal{K}_Y(Y_k, Y_\ell) - \frac{1}{M} \sum_{m=1}^M \mathcal{K}_Y(Y_k, \widehat{Y}_\ell^{(m)}) - \frac{1}{M} \sum_{m=1}^M \mathcal{K}_Y(Y_\ell, \widehat{Y}_k^{(m)}) \\ &\quad + \frac{1}{M^2} \sum_{m_1, m_2=1}^M \mathcal{K}_Y(\widehat{Y}_k^{(m_1)}, \widehat{Y}_\ell^{(m_2)}).\end{aligned}$$

In Section 3, we will show that under H_0 and suitable assumptions, $n\widehat{T}_J$ converges in distribution to the sum of J independent variables, each of which follows a typical limiting distribution of a degenerate U-statistic. Moreover, thanks to the double robustness property, the test based on $\widehat{T}_J$ will have (asymptotically) accurate size if the conditional generators have error decay exponents α_1 and α_2 such that $\alpha_1 + \alpha_2 > 1/2$ for some specific distance measures D_X and D_Y . This allows for non-parametric rates for conditional mean embedding estimation.

2.4.1 A wild bootstrap procedure for test calibration

The limiting null distribution of $n\widehat{T}_J$ is not pivotal and involves unknown quantities related to eigenvalues of certain operators depending on the kernels and the unknown joint distribution P_{XYZ} ; see Theorem 1 for details. These unknown quantities make the limiting null distribution (and its quantiles) difficult to approximate in practice. Therefore, we instead adopt a wild bootstrap procedure (see, e.g. Section 2.4 of X. Zhang et al. (2018)) to approximate the null distribution of $n\widehat{T}_J$. More concretely, we use B to denote the bootstrap sample size, and for each $b = 1, 2, \dots, B$, we generate n i.i.d. random multipliers $\{e_{bi}\}_{i=1}^n$ from the standard normal distribution $N(0, 1)$. A bootstrap version of $\widehat{T}_J$ is then defined by injecting the random multipliers as

$$\widehat{T}_J^b = \frac{1}{J} \sum_{j=1}^J \left\{ \frac{1}{\frac{n}{J} \left(\frac{n}{J} - 1 \right)} \sum_{\substack{k \neq \ell \\ k, \ell \in \mathcal{J}^{(j)}}} \widehat{U}(X_k, X_\ell) \widehat{V}(Y_k, Y_\ell) \mathcal{K}_Z(Z_k, Z_\ell) e_{bk} e_{b\ell} \right\}, \quad (12)$$

where $\widehat{U}(X_k, X_\ell)$ and $\widehat{V}(Y_k, Y_\ell)$ are defined as earlier. We then reject H_0 at level $\gamma \in (0, 1)$ if the empirical bootstrap reject proportion given the data $\mathcal{D}$ falls below γ , or $B^{-1} \sum_{b=1}^B 1_{\{\widehat{T}_J^b > \widehat{T}_J\}} < \gamma$.

3 Theoretical properties

In this section, we provide asymptotic analysis of the proposed plug-in test statistic $\widehat{T}_J$ based on the improved oracle statistic T^* using sample splitting and cross-fitting, as defined in Section 2.4. Our asymptotic analysis includes deriving the limiting distribution of the test statistic and proving the bootstrap consistency under the null H_0 as well as analysing the power of the corresponding testing procedure under a sequence of local alternatives $\{H_{1n}\}_{n=1}^\infty$ that lie near and/or above the detection boundary.

3.1 Limiting distribution and bootstrap consistency under H_0

Before presenting our theoretical results, we need to make the following assumption on the employed kernels and qualities of conditional generators $\{(\widehat{\mathbb{G}}_X^{(j)}, \widehat{\mathbb{G}}_Y^{(j)})\}_{j=1}^J$, implicitly through their generated samples $(\widehat{X}_i^{(m)}, \widehat{Y}_i^{(m)})$ for $i \in [n]$ and $m \in [M]$, trained during sampling splitting. Recall that

$\tilde{X} = \mathbb{G}_X^*(\eta, Z)$ and $\tilde{Y} = \mathbb{G}_Y^*(\kappa, Z)$ denote a valid pair of random vectors satisfying all properties in Lemma 1, whose marginal conditional distributions given Z are $P_{X|Z}$ and $P_{Y|Z}$ respectively and we denote $\tilde{X}_i^{(m)} = \mathbb{G}_X^*(\eta_i^m, Z_i)$, $\tilde{Y}_i^{(m)} = \mathbb{G}_Y^*(\kappa_i^m, Z_i)$.

Assumption 2 Assume $M \rightarrow \infty$ as $n \rightarrow \infty$. There exist constants $C_0 > 0$, $\alpha_1, \alpha_2 \geq 0$ satisfying $\alpha_1 + \alpha_2 > \frac{1}{2}$, such that for $D_i \in \{X_i, \tilde{X}_i^{(1)}, \hat{X}_i^{(1)}\}$, $E_i \in \{Y_i, \tilde{Y}_i^{(1)}, \hat{Y}_i^{(1)}\}$ and $i \in \{i_1, \dots, i_J\}$ where $i_s \in \mathcal{J}^{(s)}$ for each $s \in [J]$, we have:

$$\begin{aligned} \text{(a)} \quad & \mathbb{E}[\mathcal{K}_X(D_i, D_i) \mathcal{K}_Y(E_i, E_i) \mathcal{K}_Z(Z_i, Z_i)] < C_0. \\ \text{(b)} \quad & \sqrt{\mathbb{E}[\|\mathbb{E}[\mathcal{K}_X(\cdot, X_i) | Z_i] - \mathbb{E}[\mathcal{K}_X(\cdot, \hat{X}_i^{(1)}) | Z_i]\|_{\mathbb{H}_X}^2 \cdot (\sqrt{\mathcal{K}_Z(Z_i, Z_i)} \\ & \quad \sqrt{+\mathcal{K}_Y(E_i, E_i) \mathcal{K}_Z(Z_i, Z_i)})]} = O(n^{-\alpha_1}) \text{ and} \\ & \sqrt{\mathbb{E}[\|\mathbb{E}[\mathcal{K}_Y(\cdot, Y_i) | Z_i] - \mathbb{E}[\mathcal{K}_Y(\cdot, \hat{Y}_i^{(1)}) | Z_i]\|_{\mathbb{H}_Y}^2 \cdot (\sqrt{\mathcal{K}_Z(Z_i, Z_i)} \\ & \quad \sqrt{+\mathcal{K}_X(D_i, D_i) \mathcal{K}_Z(Z_i, Z_i)})]} = O(n^{-\alpha_2}). \end{aligned}$$

We make two remarks regarding this assumption.

Remark 2 If $\mathcal{K}_X, \mathcal{K}_Y, \mathcal{K}_Z$ are bounded kernels (without loss of generality, assume they are bounded by 1), then Assumption 2 reduces to $\mathbb{E}[\|\mathbb{E}[\mathcal{K}_X(\cdot, X_i) | Z_i] - \mathbb{E}[\mathcal{K}_X(\cdot, \hat{X}_i^{(1)}) | Z_i]\|_{\mathbb{H}_X}^2] = O(n^{-2\alpha_1})$ and $\mathbb{E}[\|\mathbb{E}[\mathcal{K}_Y(\cdot, Y_i) | Z_i] - \mathbb{E}[\mathcal{K}_Y(\cdot, \hat{Y}_i^{(1)}) | Z_i]\|_{\mathbb{H}_Y}^2] = O(n^{-2\alpha_2})$. Note that

$$\begin{aligned} & \mathbb{E}\left[\left\|\mathbb{E}[\mathcal{K}_X(\cdot, X_i) | Z_i] - \mathbb{E}[\mathcal{K}_X(\cdot, \hat{X}_i^{(1)}) | Z_i]\right\|_{\mathbb{H}_X}^2\right] \\ &= \mathbb{E}\left[\left\{\sup_{f \in \mathbb{H}_X : \|f\|_{\mathbb{H}_X} \leq 1} \mathbb{E}\left[f(\hat{X}_i^{(1)}) - f(X_i) | Z_i\right]\right\}^2\right] \\ &\leq \mathbb{E}\left[\left\{\sup_{f : \mathbb{R}^{d_X} \rightarrow \mathbb{R} : \|f\|_{\infty} \leq 1} \mathbb{E}\left[f(\hat{X}_i^{(1)}) - f(X_i) | Z_i\right]\right\}^2\right] = 2\mathbb{E}\left[d_{\text{TV}}^2(P_{\hat{X}_i^{(1)}|Z_i}, P_{X_i|Z_i})\right], \end{aligned} \tag{13}$$

where $\|\cdot\|_{\infty}$ denotes the function supreme norm, and $d_{\text{TV}}(\cdot, \cdot)$ denotes the total variation distance. Here, the inequality in the second line is implied by the fact that $\|f\|_{\infty} = \sup_{x \in \mathbb{R}^{d_X}} |f(x)| = \sup_{x \in \mathbb{R}^{d_X}} |\langle f, \mathcal{K}_X(x, \cdot) \rangle_{\mathbb{H}_X}| \leq \|f\|_{\mathbb{H}_X} \sqrt{\mathcal{K}_X(x, x)} \leq \|f\|_{\mathbb{H}_X}$. Therefore, we can also replace the error metric in Assumption 2 from an MMD type distance to the total variation distance, i.e. $\mathbb{E}[d_{\text{TV}}^2(P_{\hat{X}_i^{(1)}|Z_i}, P_{X_i|Z_i})] = O(n^{-2\alpha_1})$ and $\mathbb{E}[d_{\text{TV}}^2(P_{\hat{Y}_i^{(1)}|Z_i}, P_{Y_i|Z_i})] = O(n^{-2\alpha_2})$,

which is a common assumption made in existing works for characterizing qualities of conditional generators, see X. Zhou et al. (2022) and Shi et al. (2021). However, the total variation metric may not be a suitable metric for characterizing the closeness between nearly mutually singular distributions, which happens when data are complex objects such as images or texts exhibiting low-dimensional manifold structures (Tang & Yang, 2023). In addition, in our theory, we only require the Monte Carlo sample size M for approximating the conditional expectations to diverge to infinity as $n \rightarrow \infty$, while existing works such as Shi et al. (2021) require M to be at least proportional to n .

Remark 3 The conditional generator $\hat{\mathbb{G}}_X^{(j)}$ trained via the GMMN framework as described in Section 2.3.3 can be viewed as the minimizer of an empirical risk associated with the squared MMD between P_{XZ} and $\hat{P}_{XZ}$ defined as

$$\begin{aligned} & \left\| \mathbb{E}[\mathcal{K}_X(X_i, \cdot) \mathcal{K}_Z(Z_i, \cdot)] - \mathbb{E}[\mathcal{K}_X(\widehat{X}_i^{(1)}, \cdot) \mathcal{K}_Z(Z_i, \cdot)] \right\|_{\mathbb{H}_{XZ}}^2 \\ &= \left\{ \sup_{f \in \mathbb{H}_X \otimes \mathbb{H}_Z : \|f\|_{\mathbb{H}_{XZ}} \leq 1} \mathbb{E} \left[f(\widehat{X}_i^{(1)}, Z) - f(X_i, Z) \right]^2 \right\}, \end{aligned}$$

where $\|\cdot\|_{\mathbb{H}_{XZ}}$ denotes the RKHS norm associated with kernel $\mathcal{K}_X \otimes \mathcal{K}_Z$. By comparing this representation with the first equality in (13), we see that the population-level risk of GMMN is closely connected to and slightly weaker than the distance measure we used in our assumption for characterizing the conditional generator estimation error. The essential difference is that the supremum in our distance measure is inside the expectation. In fact, it can be shown that the quantity in the preceding display is always upper bounded by

$$\mathbb{E} \left[\left\| \mathbb{E}[\mathcal{K}_X(\cdot, X_i) | Z_i] - \mathbb{E}[\mathcal{K}_X(\cdot, \widehat{X}_i^{(1)}) | Z_i] \right\|_{\mathbb{H}_X}^2 \cdot \mathcal{K}_Z(Z_i, Z_i) \right],$$

which follows from a generalized Jensen's inequality (Park & Muandet, 2020, Appendix A). For translation invariant kernels, the above quantity is exactly the error metric we used in part (b) of Assumption 2. We expect that estimation error of the conditional generator estimator $\widehat{\mathbb{G}}_X^{(j)}$ based on the GMMN relative to this slightly stronger error metric is also possible to control; however, its analysis is beyond the scope of the current work and is left for future study. In addition, our empirical results in Appendix E, online supplementary material show that conditional generator trained using the GMMN framework with a matching MMD improves the quality of the resulting testing procedure when compared to conditional generator trained via other framework (e.g. GANs with Sinkhorn loss) in terms of size control.

The following theorem provides the limiting null distribution of our statistic, whose proof is provided in Appendix I.1, online supplementary material.

Theorem 1 Suppose Assumption 2 holds, then under H_0 , as $n \rightarrow \infty$ and $M \rightarrow \infty$, we have

$$\begin{aligned} \widehat{T}_J - T_J &= O_p(n^{-1}[M^{-1/2} + n^{-\alpha_1} + n^{-\alpha_2}] + n^{-1/2-(\alpha_1+\alpha_2)}) \quad \text{and} \\ n\widehat{T}_J &\xrightarrow{D} T^\dagger = \sum_{j=1}^J T_j^\dagger, \end{aligned}$$

where T_J is defined in the same way as $\widehat{T}_J$ with $\widehat{U}(X_j, X_k)$, $\widehat{V}(Y_j, Y_k)$ replaced by $U(X_j, X_k)$, $V(Y_j, Y_k)$ (see Equation (A-7) in Appendix I, online supplementary material), and $\{T_j^\dagger\}_{j \in [J]}$ are i.i.d. random variables with $T_1^\dagger = \sum_{s=1}^\infty \lambda_s(\chi_s^2 - 1)$. Here, χ_s^2 are i.i.d. chi-square random variables with one degree of freedom, and λ_s 's are eigenvalues of the compact self-adjoint operator on $L_2(\mathbb{R}^{d_X+d_Y+d_Z}, P_{XYZ})$ induced by the kernel function $b((X_1, Y_1, Z_1), (X_2, Y_2, Z_2)) = U(X_1, X_2)V(Y_1, Y_2)\mathcal{K}_Z(Z_1, Z_2)$; that is, there exists orthonormal basis $\{f_i(X_1, Y_1, Z_1)\}_{i=1}^\infty$ of $L_2(\mathbb{R}^{d_X+d_Y+d_Z}, P_{XYZ})$ such that

$$\begin{aligned} & \int b((X_1, Y_1, Z_1), (X_2, Y_2, Z_2)) f_i(X_1, Y_1, Z_1) dP_{XYZ}(X_1, Y_1, Z_1) \\ &= \lambda_i f_i(X_2, Y_2, Z_2). \end{aligned}$$

Remark 4 The proof of Theorem 1 depends on a crucial lemma in Appendix J, online supplementary material, which shows the asymptotic order of the difference between $\widehat{T}_J$ and the oracle statistic T_J . Since $\widehat{T}_J - T_J = o_p(n^{-1})$ if $\alpha_1 > 0$, $\alpha_2 > 0$, $M \rightarrow \infty$

and $\alpha_1 + \alpha_2 > 1/2$, under Assumption 2 we only need to show $nT_J \xrightarrow{D} T^\dagger$. Note that $T^\dagger$ is not pivotal and contains unknown quantities that are hard to estimate, which motivates the wild bootstrap procedure.

Let $\{e_i\}_{i=1}^n$ be a sequence of independent standard normal random variables, we define $\widehat{M}_J^{nM}$ according to Equation (12) with $\{e_{bi}\}_{i=1}^n$ replaced by $\{e_i\}_{i=1}^n$. For a statistic B_{nM} that depends on $\{X_i, Y_i, Z_i, \eta_i^m, \kappa_i^m\}_{i=1, m=1}^{n, M}$, as in Definition 2.1 of X. Zhang et al. (2018), we say B_{nM} converges in distribution in probability to a random variable B^* if for any subsequence $B_{n_k M_k}$, there is a further subsequence $B_{n_{k_j} M_{k_j}}$ such that $B_{n_{k_j} M_{k_j}} \{X_i, Y_i, Z_i, \eta_i^m, \kappa_i^m\}_{i, m=1}^\infty$ converges in distribution to B^* for a.e. $\{X_i, Y_i, Z_i, \eta_i^m, \kappa_i^m\}_{i, m=1}^\infty$. Let $\xrightarrow{D^*}$ denote convergence in distribution in probability. We have the following theorem on bootstrap consistency under H_0 , which is proved in Appendix I.3, online supplementary material.

Theorem 2 Suppose Assumption 2 holds, then we have $n\widehat{M}_J^{nM} \xrightarrow{D^*} T^\dagger$ as $n \rightarrow \infty$ under H_0 .

This theorem shows that the bootstrapping test statistic $\widehat{M}_J^{nM}$ admits the same limiting distribution (given data) after appropriate rescaling as the one in Theorem 2 for $\widehat{T}_J$. Therefore, the rejection threshold for $\widehat{T}_J$ computed as the (conditional) quantile based on $\widehat{M}_J^{nM}$ will lead to an asymptotically valid size for the CI test.

3.2 Power analysis under local alternatives

To facilitate the power analysis and characterize the conditional dependence detection capability of the proposed testing procedure, we introduce a sequence of local alternatives $\{H_{1n}\}_{n=1}^\infty$ based on perturbing the conditional mean embedding of the null H_0 . Recall that the triple under H_0 , denoted as (X^0, Y^0, Z^0) to distinguish it from the triple under local alternatives, satisfies $X^0 \perp\!\!\!\perp Y^0 \mid Z^0$. We consider a sequence of alternative hypothesis consisting of triple (X_n^A, Y^0, Z^0) such that $X_n^A \not\perp\!\!\!\perp Y^0 \mid Z^0$ and certain difference measure between the mean embeddings of $P_{X_n^A \mid Y^0, Z^0}$ and $P_{X_n^A \mid Z^0}$ converges to zero at rate $n^{-\alpha}$ for some index $\alpha \geq 0$; for example, $\alpha = 0$ corresponds to a fixed alternative where $X_n^A = X^A$ does not change with n .

Since (Y^0, Z^0) in the local alternative H_{1n} is the same as the null, we only need to specify the conditional distribution $P_{X_n^A \mid Y^0, Z^0}$ of X_n^A given (Y^0, Z^0) . Under Assumption 1, the conditional mean embedding of X given (Y, Z) uniquely determines $P_{X \mid YZ}$, it is therefore more convenient to specify H_{1n} through the conditional mean embedding $\mathbb{E}[K_X(X_n^A, \cdot) \mid Y^0, Z^0] \in \mathbb{H}_X$ as follows,

$$H_{1n} : \mathbb{E}[K_X(X_n^A, \cdot) \mid Y^0, Z^0] = \mathbb{E}[K_X(X^0, \cdot) \mid Z^0] + n^{-\alpha} \mathcal{G}(Y^0, Z^0), \quad (14)$$

where $\mathcal{G} : \mathbb{R}^{d_Y} \times \mathbb{R}^{d_Z} \rightarrow \mathbb{H}_X$ is any fixed $\mathbb{H}_X$ -valued mapping (not changing with n). Note that under the null H_0 , we have $\mathbb{E}[K_X(X^0, \cdot) \mid Y^0, Z^0] = \mathbb{E}[K_X(X^0, \cdot) \mid Z^0]$, corresponding to $\mathcal{G} \equiv 0$ or the limit of H_{1n} as $n \rightarrow \infty$ when $\alpha > 0$. Moreover, further restricting $\mathbb{E}[\mathcal{G}(Y^0, Z^0) \mid Z^0] = 0$ can preserve the same (conditional) marginal distribution of X_n^A given Z^0 as that under the null, that is, $P_{X_n^A \mid Z^0} = P_{X^0 \mid Z^0}$, which, however, is not necessary. Under H_{1n} , we have

$$\text{MMDCI}(P_{X_n^A Y^0 Z^0}) = n^{-2\alpha} \mathbb{E} \left[\left\langle \mathcal{G}(Y^0, Z^0), \mathcal{G}(\tilde{Y}^0, \tilde{Z}^0) \right\rangle_{\mathbb{H}_X} \cdot V(Y^0, \tilde{Y}^0) \cdot K_Z(Z^0, \tilde{Z}^0) \right] = O(n^{-2\alpha}),$$

where $\tilde{Y}^0$ and $\tilde{Z}^0$ are independent copies of Y^0 and Z^0 , respectively. Note that $\text{MMDCI}(P_{X_n^A Y^0 Z^0})$ is the squared MMD (relative to the kernel $K_0 = K_X \otimes K_Y \otimes K_Z$) between the joint distributions $P_{X_n^A Y^0 Z^0}$ and $P_{\tilde{X}_n^A Y^0 Z^0}$, where $\tilde{X}_n^A$ satisfies that $\tilde{X}_n^A \perp\!\!\!\perp Y^0 \mid Z^0$ and $(\tilde{X}_n^A, Z^0) \stackrel{d}{=} (X_n^A, Z^0)$ so that $(\tilde{X}_n^A, Y^0, Z^0)$ satisfies the null H_0 . Therefore, the rate $n^{-\alpha}$ quantifies the degree of deviation in the local alternative H_{1n} from the null H_0 as $n \rightarrow \infty$. In a related work, Cai et al. (2022) considered specifying a local alternative $P_{X_n^A \mid Y^0, Z^0} = P_{X^0 \mid Z^0} + n^{-1/2} (P_{X^A \mid Y^0, Z^0} - P_{X^A \mid Z^0})$ based on directly perturbing the conditional distribution $P_{X^0 \mid Y^0, Z^0}$ of X^0 given (Y^0, Z^0) , which can be viewed as a

special case of our characterization of local alternatives with $\mathcal{G}(Y^0, Z^0) = \int_{\mathbb{H}_X} \mathcal{K}_X(x, \cdot) [dP_{X^A|Y^0, Z^0}(x) - dP_{X^A|Z^0}(x)] \in \mathbb{H}_X$ and $\alpha = 1/2$.

For any fixed n , let $\{(X_i, Y_i, Z_i)\}_{i=1}^n$ be i.i.d copies of (X_n^A, Y^0, Z^0) . The following theorem gives the asymptotic properties of our statistic under H_{1n} , which is proved in [Appendix I.2, online supplementary material](#).

Theorem 3 Suppose Assumptions 1 and 2 hold, then under H_{1n} , as $n \rightarrow \infty$:

1. If $\alpha = 0$, then $\sqrt{n}(\widehat{T}_J - c_0) \xrightarrow{D} \frac{1}{\sqrt{J}} \sum_{j=1}^J \mathcal{G}_j^{(0)}$, where $c_0 = \mathbb{E}\{U(X_1, X_2) V(Y_1, Y_2) \mathcal{K}_Z(Z_1, Z_2)\} > 0$ and $\mathcal{G}_j^{(0)}$ are independent mean zero normal random variables with variance equal to

$$4\text{Var}\left(\mathbb{E}\left[U(X_1, X_2) V(Y_1, Y_2) \mathcal{K}_Z(Z_1, Z_2) \middle| X_2, Y_2, Z_2\right]\right).$$

2. If $0 < \alpha < 1/2$, then $n^{2\alpha} \widehat{T}_J \xrightarrow{p} c$, where $c = \mathbb{E}\{\langle \mathcal{G}(Y_1, Z_1), \mathcal{G}(Y_2, Z_2) \rangle_{\mathbb{H}_X} V(Y_1, Y_2) \mathcal{K}_Z(Z_1, Z_2)\} > 0$.
3. If $\alpha = 1/2$, then $n \widehat{T}_J \xrightarrow{D} c + T^\dagger + \frac{1}{\sqrt{J}} \sum_{j=1}^J \mathcal{G}_j$, where $\mathcal{G}_j$ are independent mean zero normal random variables, possibly correlated with $T_j^\dagger$ in Theorem 1, with variance equal to

$$4\text{Var}\left(\mathbb{E}\left[\langle \mathcal{G}(Y, Z), \mathcal{K}_X(X', \cdot) - \mathbb{E}[\mathcal{K}_X(X', \cdot) | Z'] \rangle_{\mathbb{H}_X} + \langle \mathcal{G}(Y', Z'), \mathcal{K}_X(X, \cdot) - \mathbb{E}[\mathcal{K}_X(X, \cdot) | Z] \rangle_{\mathbb{H}_X} \cdot V(Y, Y') \mathcal{K}_Z(Z, Z') \middle| X', Y', Z'\right]\right),$$

where (X, Y, Z) and (X', Y', Z') are i.i.d copies from $P_{X^0 Y^0 Z^0}$.

4. If $\alpha > 1/2$, $n \widehat{T}_J \xrightarrow{D} T^\dagger$.

The following theorem shows the asymptotic behaviour of the bootstrapping statistic $\widehat{M}_J^{nM}$, as well as the asymptotic power of our proposed testing procedure, under H_{1n} , which is proved in [Appendix I.3, online supplementary material](#).

Theorem 4 Suppose Assumption 2 holds, then we have that as $n \rightarrow \infty$ and $M \rightarrow \infty$:

1. Under H_{1n} with $\alpha > 0$, $n \widehat{M}_J^{nM} \xrightarrow{D^*} T^\dagger$.
2. Under H_{1n} with $\alpha = 0$, $n \widehat{M}_J^{nM} \xrightarrow{D^*} T_1 = \sum_{j=1}^J \widetilde{T}_j$, where $\widetilde{T}_j$ are i.i.d random variables with $\widetilde{T}_1 = \sum_{i=1}^\infty \gamma_i (\chi_i^2 - 1)$, χ_i^2 are i.i.d chi-square random variables with one degree of freedom and γ_i s are eigenvalues of $b((X_1, Y_1, Z_1), (X_2, Y_2, Z_2)) = U(X_1, X_2) V(Y_1, Y_2) \mathcal{K}_Z(Z_1, Z_2)$. If we further assume Assumption 1 holds and let $M_{nM, \gamma}^*$ denote the $(1 - \gamma)$ th quantile of $n \widehat{M}_J^{nM}$ conditioning on $\{X_i, Y_i, Z_i, \eta_i^m, \kappa_i^m\}_{i=1, m=1}^{n, M}$, then under H_{1n} : $P(n \widehat{T}_J \geq M_{nM, \gamma}^*) \rightarrow 1$ if $\alpha < 1/2$, $P(n \widehat{T}_J \geq M_{nM, \gamma}^*) \rightarrow P(c + T^\dagger + \frac{1}{\sqrt{J}} \sum_{j=1}^J \mathcal{G}_j \geq T_{0, \gamma})$ if $\alpha = 1/2$ and $P(n \widehat{T}_J \geq M_{nM, \gamma}^*) \rightarrow \gamma$ if $\alpha > 1/2$.

According to Theorems 1 and 4, our proposed testing procedure asymptotically achieves the correct size under H_0 and is consistent in the sense that it can correctly reject H_0 under any (local) alternative H_1 lying outside a $n^{-1/2}$ -neighbourhood of H_0 . This means that our test is shown to have nontrivial power under alternatives that approach H_0 at the rate $n^{-1/2}$, while in a related work ([Pogodin et al., 2024](#)) based on kernel smoothing, the consistency of their method is only derived under a fixed alternative.

4 Simulation study

In this section, we examine the size and power properties of our proposed test statistic and compare with several existing tests. In Section 4.1, we investigate the influence of the fold number J on the empirical performance of $\hat{T}_J$. In Section 4.2, we consider a post nonlinear noise model and examine the performance of $\hat{T}_2$ (i.e. take $J=2$) when the dimension d_Z of Z is large. Section 4.3 contains the result for a weakly conditional independent alternative where both GCIT proposed in Bellot and der Schaar (2019) and DGCIT proposed in Shi et al. (2021) have trivial power. Under the alternative, the size-adjusted power for all bootstrap-based methods examined in this section are calculated according to Domínguez and Lobato (2000). For all the simulations in Section 4 and the two applications in Section 5, we opt to use the Laplacian kernel. The bandwidth parameter for each kernel is selected according to the median heuristic (Gretton et al., 2012, Section 8). For example, the bandwidth parameter for j th fold is defined as $\sigma_j^X = \text{median}\{\|X_i - X_s\|_1 : i, s \in \mathcal{J}^{(j)}\}$ for $j \in [J]$. We remark that our empirical result of $\hat{T}_J$ is not overly sensitive to the number of synthetic data M and the number of bootstrap replicates B , so we simply fix $M = 100$ and $B = 1,000$.

4.1 Sensitivity with respect to the fold number J

We investigate the impact of fold number J on the performance of $\hat{T}_J$ with $J \in \{2, 4, 8\}$. We consider a data generation process (DGP), where $Z_i = e_{3i}$, $Y_i = Z_i + e_{1i}$, $X_i = Z_i + \delta_i e_{1i} + (1 - \delta_i) e_{2i}$ with $\{e_{1i}\}_{i=1}^n$, $\{e_{2i}\}_{i=1}^n$, and $\{e_{3i}\}_{i=1}^n$ being independent samples following the standard normal distribution and $\delta_i \stackrel{i.i.d.}{\sim} \text{Bernoulli}(p)$, which is also independent of $\{e_{si}\}_{i=1}^n$ for $s \in [3]$. The experiments are repeated 1,000 times with nominal level being fixed at 5%.

For the empirical size, we fix $p = 0$ and set $n \in \{200, 400, 600, 800, 1,000\}$. As shown in Figure 1a, the empirical size for $\hat{T}_J$ is close to the nominal level for all $J \in \{2, 4, 8\}$ for all sample size n . For the size-adjusted power, we set $n = 400$ and plot the power curves against $p \in [0, 0.25]$ in Figure 1b. For any fixed p , the empirical power for $\hat{T}_J$ decreases as J increases. This is expected since only the $J \times \frac{n^2}{J}$ entries in the J block diagonal distance matrices of $\{\mathcal{K}_X(X_j, X_k)\}_{j,k=1}^n$, $\{\mathcal{K}_Y(Y_j, Y_k)\}_{j,k=1}^n$, and $\{\mathcal{K}_Z(Z_j, Z_k)\}_{j,k=1}^n$ are used in evaluating $\hat{T}_J$ and the number of pairs declines as J increases. Since the empirical size is not sensitive to the choice of J and smaller value of J corresponds to larger power, we recommend setting $J = 2$ in practice.

4.2 Post nonlinear noise model

Consider the following DGP adopted by Bellot and der Schaar (2019) and Shi et al. (2021): $Y_i = \sin(a_f^T Z_i + e_{fi})$, $X_i = \cos(a_g^T Z_i + b Y_i + e_{gi})$ and $Z_i \stackrel{i.i.d.}{\sim} \mathcal{N}(0, I_{d_Z})$, where $\{e_{fi}\}_{i=1}^n$, $\{e_{gi}\}_{i=1}^n$ are independent samples from $\mathcal{N}(0, 0.25)$ and the entries of $a_f, a_g \in \mathbb{R}^{d_Z}$ are normalized to the unit ℓ_1 norm after randomly sampled from Uniform $([0, 1]^{d_Z})$. As comparison, we also include the simulation results for DGCIT, GCIT, the CIT test proposed in Cai et al. (2022) (denoted as CIT) and an oracle statistic $\hat{T}_2$ oracle, which is $\hat{T}_2$ with synthetic data generated from the true conditional generators. The experiments are repeated 500 times with sample size $n = 2,000$ and nominal level being fixed at 5%.

For the empirical size, we fix $b = 0$ (which corresponds to H_0) and set $d_Z \in \{50, 100, 150, 200, 250\}$. As shown in Figure 2a, $\hat{T}_2$ and $\hat{T}_2$ oracle have relatively accurate empirical size for $d_Z \leq 200$ and $\hat{T}_2$ is mildly undersized when $d_Z = 250$. GCIT is undersized when $d_Z \leq 150$ and is oversized when $d_Z = 250$. DGCIT goes from oversized to undersized as d_Z increases at 5% level and it has large size distortion when $d_Z = 150$ at level 10%. For CIT, the empirical sizes are close to the nominal levels for all values of d_Z . However, CIT is much more computationally expensive than $\hat{T}_2$. As shown in Table 1, it takes 18 min for CIT to finish one Monte Carlo experiment on an Intel Core i7-11800H CPU when $d_Z = 50$ and the time increases to 107 min when $d_Z = 250$. For $\hat{T}_2$ the time for one experiment is about 3 min for all values of d_Z .

For the empirical power, we set $d_Z = 200$ and plot the power curves against $b \in [0.15, 0.75]$ in Figure 2b. In general, $\hat{T}_2$ has the largest power compared with DGCIT, GCIT, and CIT for all values of b and its power curve is close to $\hat{T}_2$ oracle. DGCIT outperforms GCIT when $b \leq 0.45$ while the latter has better performance when $b \in [0.6, 0.75]$. CIT has almost trivial power for

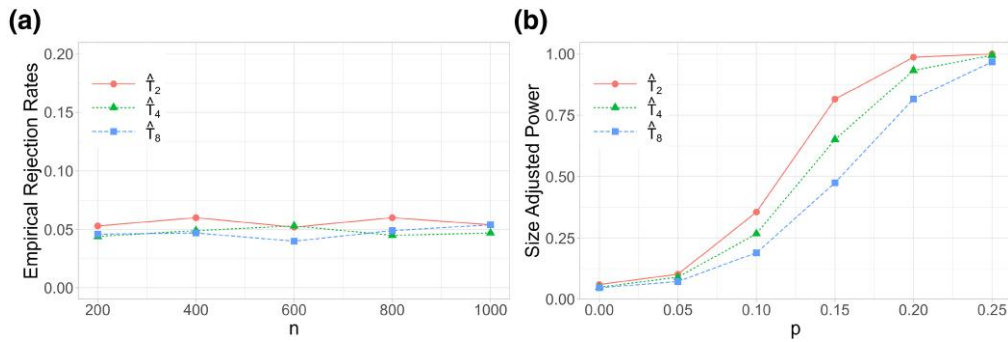


Figure 1. Empirical size (left) and size-adjusted power (right) for $\hat{T}_J$.

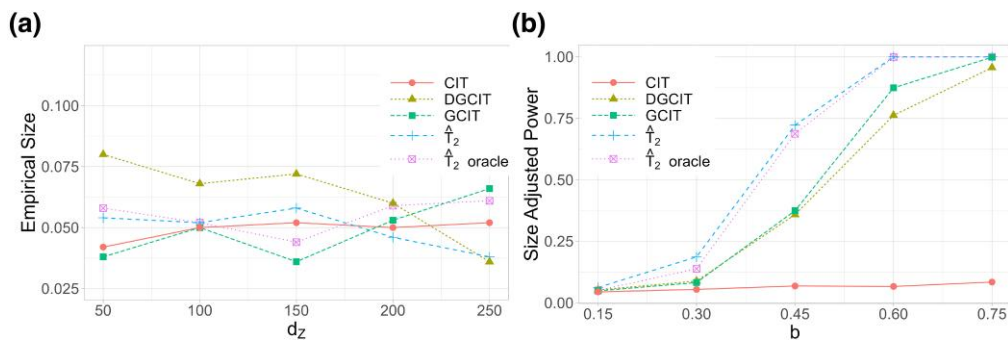


Figure 2. Empirical size (left) and size-adjusted power (right) at nominal level 5%.

Table 1. Computation time for one Monte Carlo experiment (in minutes). All experiments (except CIT) were conducted on Google Colab using the NVIDIA Tesla T4 GPU. CIT was run on an Intel Core i7-11800H CPU. GCIT used an early stopping criterion.

	d_Z				
	50	100	150	200	250
GCIT	0.50	0.50	0.50	0.50	0.50
$\hat{T}_2$	2.83	3.00	3.05	3.23	3.3
DGCIT	4.85	5.33	5.67	6.23	6.45
CIT	18.33	41.27	56.32	80.57	107.35

all values of b . This is expected since it requires iterative estimation of conditional cumulative distribution functions (Cai et al., 2022, Section 3.1), which suffers from the curse of dimensionality and yields poor power performance when the dimension of the conditioning variable is large.

4.3 Weakly conditional independent alternative

In this subsection, we show that both DGCIT and GCIT have trivial power against the alternative which is only weakly conditionally independent. Under H_0 , we generate $\{X_i\}_{i=1}^n$, $\{Y_i\}_{i=1}^n$, and $\{Z_i\}_{i=1}^n$ as mutually independent samples from Bernoulli(0.5). Under the alternative, we consider the same

Table 2. Empirical rejection rate under null and weakly conditional independent alternative

	n	Null						Alternative					
		$\hat{T}_2$		DGCIT		GCIT		$\hat{T}_2$		DGCIT		GCIT	
		10%	5%	10%	5%	10%	5%	10%	5%	10%	5%	10%	5%
Oracle	200	10.4	4.3	11.5	6.3	10.3	5.2	96.8	91.6	13.5	7.2	8.3	4
	600	10.7	5.5	11.2	6.3	8.2	4.3	100	100	8.5	5	10.1	4.9
	1,000	9.9	3.9	9.9	4.3	8.2	4.0	100	100	9.8	4.3	10.1	5
Practical	200	10.6	4.8	11.2	6.6	12.8	6.2	95.2	89	12.2	7.2	14	6.6
	600	9	4.2	11.8	6.6	10.4	4.4	100	100	12.8	7.2	11.6	5.6
	1,000	11	5.6	12	7	11.2	6.6	100	100	12	7.2	9.4	4.2

DGP as in Example 5 of [Shi et al. \(2021\)](#), where $Z_i \stackrel{i.i.d}{\sim} \text{Bernoulli}(\frac{1}{2})$ and

$$\begin{pmatrix} \Pr(X_i = 0, Y_i = 0 | Z_i = 0) & \Pr(X_i = 0, Y_i = 1 | Z_i = 0) \\ \Pr(X_i = 1, Y_i = 0 | Z_i = 0) & \Pr(X_i = 1, Y_i = 1 | Z_i = 0) \end{pmatrix} = \begin{pmatrix} 1/6 & 1/3 \\ 1/3 & 1/6 \end{pmatrix},$$

$$\begin{pmatrix} \Pr(X_i = 0, Y_i = 0 | Z_i = 1) & \Pr(X_i = 0, Y_i = 1 | Z_i = 1) \\ \Pr(X_i = 1, Y_i = 0 | Z_i = 1) & \Pr(X_i = 1, Y_i = 1 | Z_i = 1) \end{pmatrix} = \begin{pmatrix} 1/3 & 1/6 \\ 1/6 & 1/3 \end{pmatrix}.$$

Under both null and alternative hypothesis, we consider two settings: (1) the oracle setting where synthetic data are generated from the true conditional generators (with the impact of inaccuracy due to the use of estimated conditional generators being eliminated); (2) the practical setting where the conditional generators are learned from GNNs. The experiments are repeated 1,000 times with sample size $n \in \{200, 600, 1,000\}$ and nominal level being fixed at 5% and 10%. As shown in [Table 2](#), $\hat{T}_2$, DGCIT, and GCIT have accurate size under H_0 for all values of n . However, under the alternative in both oracle and practical settings, both DGCIT and GCIT have trivial power while $\hat{T}_2$ can successfully detect this alternative.

5 Real data applications

In this section, we apply our method to two real-world applications using the MNIST dataset: image classification and image denoising and deblurring, to assess its empirical performance. In [Appendix F, online supplementary material](#), we also conduct another real data example using the Cancer Cell Line Encyclopedia (CCLE) dataset ([Barretina et al., 2012](#)).

5.1 Classification with MNIST dataset

Dimension reduction is a key building block for machine learning in various fields, such as computer vision and language processing. Extracting useful low-dimensional features from complex data helps avoid the curse of dimensionality, reduce computation burden and leads to improved model performance ([Berahmand et al., 2024](#)). As a popular nonlinear feature extraction methods, the auto-encoder (AE) uses a neural network (encoder) to transform the data X to a d_l -dimensional feature vector $f_{d_l}(X)$, which is then transformed back through another neural network (decoder) for reconstruction, resulting in reconstructed data X' . The selection of the feature dimension d_l depends on the data and downstream applications and a large d_l may undermine the performance and interpretability of the model ([Bonheme & Grzes, 2022](#)).

We now demonstrate that our proposed CI test can be used to determine the best value for d_l by using the MNIST dataset that contains 70,000 images of handwritten digits $Y = 0, 1, \dots, 9$. Let $\{(X_i, Y_i)\}_{i=1}^{70,000}$ denote the 28×28 pixels of images and their corresponding digits, which are assumed to be i.i.d copies of the random vector (X, Y) . For different values of d_l , we consider testing the null hypothesis $H_0: X \perp\!\!\!\perp Y \mid f_{d_l}(X)$ against $H_1: X \not\perp\!\!\!\perp Y \mid f_{d_l}(X)$. The minimum value of d_l such

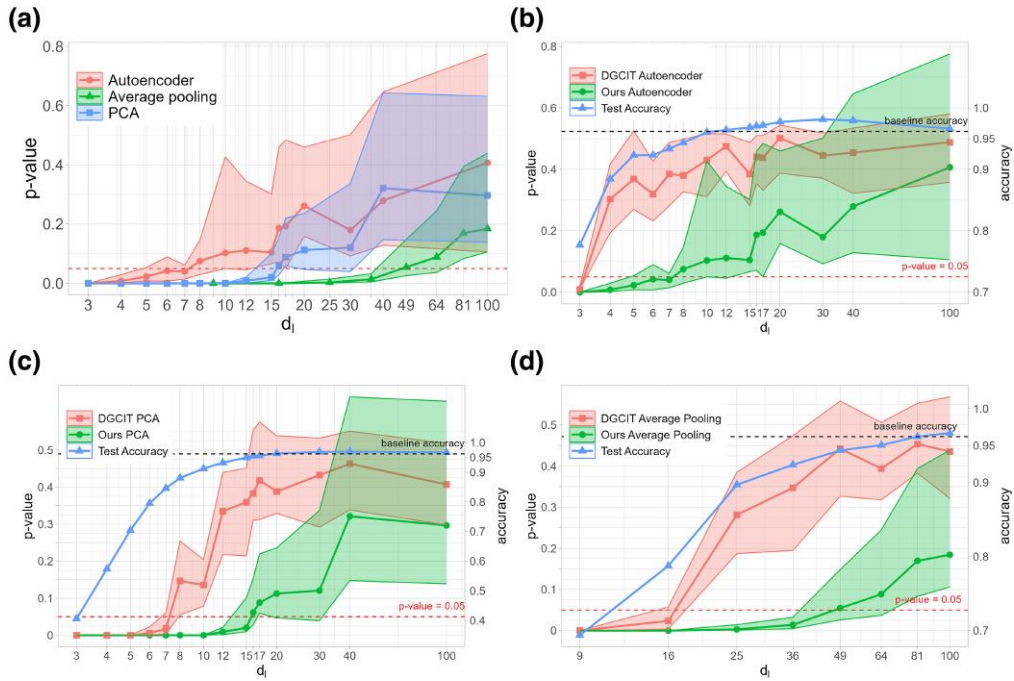


Figure 3. (a) The median p -values and its 25% and 75% quantiles for AE, average pooling, and PCA. Some sample images generated by $P_{X|f_{d_l}(X)}$ for different d_l is provided in [Appendix H.1, online supplementary material](#). (b, c, d) The test accuracies of k-NN classifiers, as well as the median p -values and its 25% and 75% quantiles of our method and DGCIT for AE, average pooling, and PCA. The baseline accuracy is obtained by training k-NN using the original images and their corresponding labels.

that H_0 is not rejected will be chosen as the dimension of the feature vector. We split the MNIST dataset into training (with size 60,000) and testing (with size 10,000) samples. Half of the images in the training dataset are used to train an AE and the other half, as well as its feature vectors from the trained encoder, is used to learn generators of $P_{Y|f_{d_l}(X)}$ and $P_{X|f_{d_l}(X)}$. The test statistic is evaluated on 40 equal folds of the testing dataset (note that we do not perform sample splitting on each fold). As a comparison, we also include the testing result when the feature vectors are extracted using average pooling (Zafar et al., 2022, Section 2.2) and principal component analysis (PCA), where all the images in the training dataset are used to learn $P_{Y|f_{d_l}(X)}$ and $P_{X|f_{d_l}(X)}$.

The median of the 40 p -values are plotted in [Figure 3a](#) as a function of d_l . For AE, the median and 25% quantile of the p -values exceeds 5% level when $d_l \geq 7$ and $d_l \geq 10$, respectively. This suggests the dimension of feature vector is about 10 to 15, which is consistent with many empirical findings for MNIST data (Fournier & Aloise, 2019). For PCA, the 75% quantile of the p -values remains close to zero when $d_l \leq 10$ and the median p -value rises above 5% level when $d_l > 15$. For average pooling, the 75% quantile of the p -values remains close to zero even if the pooled image has 5×5 pixels (implemented by `torch.nn.AdaptiveAvgPool2d` with `kernelsize = 8`, `stride = 5`, `padding = 0`). If we select the smallest value of d_l such that the 25% quantile of the p -values exceeds the 5% level as the best dimension for the feature vector, then the best dimensions selected for AE, PCA, and average pooling are 10, 17, and 64, respectively. As expected, this result demonstrates that AE is more efficient than PCA and average pooling in feature extraction.

To further demonstrate the practical applicability of our test, we train a k-NN classifier using $(f_{d_l}(X_i), Y_i)$'s on the 30,000 training samples that were not used to train AE for different values of d_l ; see [Appendix G.2.1, online supplementary material](#) for more details. The accuracy of the k-NN classifier is then evaluated on the testing samples. As shown in [Figure 3b–e](#), the best dimensions of the feature vectors selected by our CI test correspond to the smallest dimensions where the test accuracy curves begin to level off. For comparison, we also evaluated the DGCIT statistic proposed in Shi et al. (2021), whose corresponding best dimensions are 4, 8, and 25 for AE, PCA, and average pooling, respectively. These dimensions are significantly smaller than those selected by our

test, which can be attributed to the lack of power for DGCIT in high-dimensional settings. Consequently, the k-NNs trained using feature vectors at the dimensions selected by DGCIT have low and suboptimal test accuracies.

5.2 Image denoising and deblurring with MNIST dataset

For computer vision tasks such as image denoising and deblurring, object detection, and semantic segmentation, the introduction of deep learning techniques, including convolutional neural networks (CNN) (LeCun et al., 1989), has led to significant improvements in the performance and applicability of computer vision algorithms. A neural network $\mathcal{G}$ designed for these tasks typically takes an image X as input and is trained by minimizing certain loss functions between the output image $\mathcal{G}(X)$ and the target image Y . Such neural networks can be applied in many real-world scenarios, such as picture editing (Long et al., 2015) and autonomous driving (Geiger et al., 2012). For instance, in semantic segmentation applications, the input X could be images of human faces, and the target Y would be an image of the same dimension as X , with each pixel representing a class label (e.g. 1: ‘human face’ or 0: ‘background’). In this case, the neural network is trained to classify each pixel of X , so that the output $\mathcal{G}(X)$ closely matches the target Y . Another example is image denoising, which involves inspecting a noisy image and recovering an estimate of the underlying clean counterpart by eliminating noise artefacts (Izadi et al., 2023). Here, (X, Y) represents a noisy and a clear image pair of the same object.

In addition to auxiliary techniques, such as skip connections (Long et al., 2015) and residual learning (He et al., 2016), many popular neural network models for image-to-image tasks use an encoder–decoder architecture, such as U-net (Ronneberger et al., 2015), FCN (Long et al., 2015), and RED-Net (Jiang et al., 2018). The encoder part (also known as the backbone or feature extractor) consists of multiple convolution and pooling layers, where the original input image is locally convoluted and down-sampled to extract useful features (known as feature maps or channels in CNN) with gradually decreasing resolution. These feature maps are then gradually up-sampled in the decoder part and restored to output an image with the same resolution as the input. While empirical comparisons between different backbone structures already exist (Elharrouss et al., 2024; Goldblum et al., 2023), it remains unclear how to choose the dimensions of the feature maps that are fed into the decoder part of the networks. As pointed out in Bhatt et al. (2021), some of the feature maps may not significantly contribute to learning the target image Y (Kawaguchi et al., 2019), and excessive features can introduce noise, leading to overfitting and increased training costs. Moreover, choosing the smallest feature dimension can help identify the optimal image compression rate while preserving all information relevant to image reconstruction or downstream applications, thereby significantly reducing storage costs in practice.

To demonstrate the practical applicability of our test, we apply it to an image denoising and deblurring task. Using the MNIST dataset, we construct 70,000 blurred and noisy images $\{X_i\}_{i=1}^{70,000}$ from the original clear images $\{Y_i\}_{i=1}^{70,000}$ (see Appendix G.2.2, online supplementary material for details) and assume that they are i.i.d. copies of the $\mathbb{R}^{28 \times 28 \times 28}$ -valued random vector (X, Y) . We split $\{X_i, Y_i\}_{i=1}^{70,000}$ into modelling (30,000 samples), training (30,000 samples), and testing (10,000 samples) datasets. On the modelling dataset, we train multiple encoder–decoder networks $\mathcal{G}_{d_i}$ to reconstruct the clear images, where only the output feature map from the last layer of the encoder, with dimension d_i , is passed into the decoder part (all other network structures are identical for different networks). Let D_{d_i} denote the encoder part of $\mathcal{G}_{d_i}$, and $Z^{d_i} := D_{d_i}(X)$. If the dimension of Z^{d_i} is chosen correctly, it should contain all the relevant information for the decoder to reconstruct the clear image Y . We can test whether the dimension of the feature map is specified properly by testing the null hypothesis $H_0: X \perp Y|Z^{d_i}$ against the alternative $H_1: X \not\perp Y|Z^{d_i}$. Denote $Z_i = D_{d_i}(X_i)$ for any X_i in the training and testing datasets. Following a similar procedure as in Section 5.1, we learn generators of $P_{Y|Z^{d_i}}$ and $P_{X|Z^{d_i}}$ on the training dataset, and the test statistic is evaluated on 50 equal folds of the testing dataset (note that we do not perform sample splitting on each fold). For comparison, we also evaluated the DGCIT statistic proposed in Shi et al. (2021). To quantify the performance of $\mathcal{G}_{d_i}$, we calculate the peak signal-to-noise ratio (PSNR, Hore & Ziou, 2010) between $\mathcal{G}_{d_i}(X_i)$ and Y_i on the testing dataset, which is a strictly decreasing function of their pixelwise mean squared error. In addition, we also trained a k-NN classifier (following the same training procedure as the k-NN classifier used in Section 5.1) using the reconstructed images

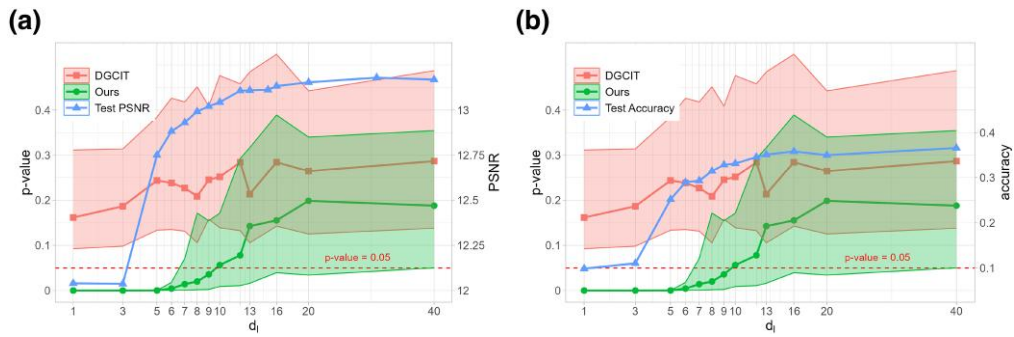


Figure 4. The median p -values and its 25% and 75% quantiles for our proposed test and DGCIT, with PSNR (a) and test accuracy of k-NN in percentage (b). The dashed line represents the 5% nominal level.

$\mathcal{G}_{d_l}(X_i)$ as well as its corresponding digits. The accuracy of the trained classifier, which is the proportion of the correctly classified samples, is evaluated on the testing dataset.

As shown in Figure 4, the median of the 50 p -values for DGCIT is significantly larger than the median p -value for our test, due to the lack of power for DGCIT when both X and Y are high-dimensional. Depending on whether the median or 25% quantile of the p -values exceeds the 5% nominal level, our test suggests a feature map dimension of 10 or 16, where the trained network $\mathcal{G}_{d_l}$ performs well according to the PSNR and test accuracy. However, DGCIT does not reject the null hypothesis at the 5% level for any values of d_l , even though the PSNR and test accuracy clearly indicate that the performance of $\mathcal{G}_{d_l}$ deteriorates drastically when $d_l \leq 5$. It is also worth mentioning that the k-NN test accuracy is 5.8% when using the blurred and noisy image X as input to learn the labelled digit. As shown in Figure 4b, the k-NN test accuracy is about 33% for $d_l \geq 10$, which we consider quite decent, given the very noisy nature of the X_i ; see Section H.2, online supplementary material for sample images. This example demonstrates the effectiveness of the trained encoder-decoder networks $\mathcal{G}_{d_l}$ and the capability of our CI-testing procedure in handling high-dimensional image data. It further showcases the potential usefulness of our tests as a diagnostic tool in the training of neural networks.

6 Conclusion

In this article, we propose a novel conditional independence (CI) test that overcomes the challenges faced by existing tests when the dimensions of the data are large. By utilizing reproducing kernel Hilbert space (RKHS) embedding, sample splitting, cross-fitting as well as generative neural networks (GNNs), the test we developed is fully non-parametric, alleviates the curse of dimensionality, and is doubly robust against approximation errors of GNN generators. Extensive simulations demonstrate the accurate size and satisfactory power performance of the proposed method in comparison with existing ones. The data examples further illustrate the versatility of our proposed method in real-world applications.

To conclude, we discuss several potential extensions. In regression modelling and sufficient dimension reduction, it is often more useful to test the weaker null hypothesis of conditional mean or quantile independence to determine whether an additional predictor contributes to predicting the mean or quantile of the response variable given a set of covariates. Generalizing the proposed test to address these problems would be beneficial. Additionally, constructing a CI test with triple or multiple robustness properties for testing mutual conditional independence among multiple random variables may be desirable. Moreover, it would be interesting to apply the current method to several downstream statistical applications. For example, in time series model specification testing, where the Markov property described through conditional independence plays an important role (Y. Zhou et al., 2023), and in nonlinear dimension reduction problems, where our test procedure could be promising in identifying the optimal dimension reduction mapping by appropriately inverting the test statistic into an estimator in a computationally efficient manner; see J. Huang et al. (2024) for some recent work. We leave these topics for further research.

- Elharrouss O., Akbari Y., Almadeed N., & Al-Madeed S. (2024). Backbones-review: Feature extractor networks for deep learning and deep reinforcement learning approaches in computer vision. *Computer Science Review*, 53(1), 100645. <https://doi.org/10.1016/j.cosrev.2024.100645>
- Fournier Q., & Aloise D. (2019). Empirical comparison between autoencoders and traditional dimensionality reduction methods. In *2019 IEEE Second International Conference on Artificial Intelligence and Knowledge Engineering (AIKE)* (pp. 211–214). IEEE.
- Fukumizu K., Gretton A., Sun X., & Schölkopf B. (2007). Kernel measures of conditional dependence. *Advances in Neural Information Processing Systems*, 20, 873–880.
- Funk M. J., Westreich D., Wiesen C., Stürmer T., Brookhart M. A., & Davidian M. (2011). Doubly robust estimation of causal effects. *American Journal of Epidemiology*, 173(7), 761–767. <https://doi.org/10.1093/aje/kwq439>
- Geiger A., Lenz P., & Urtasun R. (2012). Are we ready for autonomous driving? The kitti vision benchmark suite. In *2012 IEEE Conference on Computer Vision and Pattern Recognition* (pp. 3354–3361). IEEE.
- George E. I. (2000). The variable selection problem. *Journal of the American Statistical Association*, 95(452), 1304–1308. <https://doi.org/10.1080/01621459.2000.10474336>
- Goldblum M., Souri H., Ni R., Shu M., Prabhu V., Somepalli G., Chattopadhyay P., Ibrahim M., Bardes A., Hoffman J., Chellappa R., Wilsonhellappa A. G., & Goldstein T. (2023). Battle of the backbones: A large-scale comparison of pretrained models across computer vision tasks. *Advances in Neural Information Processing Systems*, 36, 29343–29371. <https://doi.org/10.48550/arXiv.2308.10349>
- Goodfellow I., Pouget-Abadie J., Mirza M., Xu B., Warde-Farley D., Ozair S., Courville A., & Bengio Y. (2014). Generative adversarial nets. *Advances in Neural Information Processing Systems*, 27, 2672–2680. <https://doi.org/10.48550/arXiv.1406.2661>
- Gretton A., Borgwardt K. M., Rasch M. J., Schölkopf B., & Smola A. (2012). A kernel two-sample test. *Journal of Machine Learning Research*, 13(25), 723–773. <https://doi.org/10.5555/2188385.2188410>
- Gretton A., Bousquet O., Smola A., & Schölkopf B. (2005). Measuring statistical dependence with Hilbert-Schmidt norms. In *International Conference on Algorithmic Learning Theory* (pp. 63–77). Springer.
- He K., Zhang X., Ren S., & Sun J. (2016). Deep residual learning for image recognition. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition* (pp. 770–778). IEEE Computer Society.
- Hore A., & Ziou D. (2010). Image quality metrics: PSNR vs. SSIM. In *2010 20th International Conference on Pattern Recognition* (pp. 2366–2369). IEEE.
- Huang J., Jiao Y., Liao X., Liu J., & Yu Z. (2024). Deep dimension reduction for supervised representation learning. *IEEE Transactions on Information Theory*, 70(5), 3583–3598. <https://doi.org/10.1109/TIT.2023.3340658>
- Huang Z., Deb N., & Sen B. (2022). Kernel partial correlation coefficient—A measure of conditional dependence. *Journal of Machine Learning Research*, 23(216), 1–58. <https://doi.org/10.48550/arXiv.2103.03681>
- Izadi S., Sutton D., & Hamarneh G. (2023). Image denoising in the deep learning era. *Artificial Intelligence Review*, 56(7), 5929–5974. <https://doi.org/10.1007/s10462-022-10305-2>
- Izbicki R., & Lee A. B. (2016). Nonparametric conditional density estimation in a high-dimensional regression setting. *Journal of Computational and Graphical Statistics*, 25(4), 1297–1316. <https://doi.org/10.1080/10618600.2015.1094393>
- Jiang J., Zheng L., Luo F., & Zhang Z. (2018). ‘Rednet: Residual encoder-decoder network for indoor rgb-d semantic segmentation’, arXiv, arXiv:1806.01054, preprint: not peer reviewed. <https://doi.org/10.48550/arXiv.1806.01054>
- Kallenberg O. (2002). *Foundations of modern probability* (2nd ed.). Springer.
- Kawaguchi K., Huang J., & Kaelbling L. P. (2019). Effect of depth and width on local minima in deep learning. *Neural Computation*, 31(7), 1462–1498. https://doi.org/10.1162/neco_a_01195
- Kim I., Neykov M., Balakrishnan S., & Wasserman L. (2022). Local permutation tests for conditional independence. *Annals of Statistics*, 50(6), 3388–3414. <https://doi.org/10.1214/22-AOS2233>
- Koller D., & Friedman N. (2009). *Probabilistic graphical models: Principles and techniques*. MIT Press.
- LeCun Y., Boser B., Denker J., Henderson D., Howard R., Hubbard W., & Jackel L. (1989). Handwritten digit recognition with a back-propagation network. *Advances in Neural Information Processing Systems*, 2, 396–404. <https://doi.org/10.5555/109230.109298>
- Li B. (2018). *Sufficient dimension reduction: Methods and applications with R*. Chapman and Hall/CRC.
- Li C., & Fan X. (2020). On nonparametric conditional independence tests for continuous variables. *Wiley Interdisciplinary Reviews: Computational Statistics*, 12(3), e1489. <https://doi.org/10.1002/wics.v12.3>
- Li M., Neykov M., & Balakrishnan S. (2022). Minimax optimal conditional density estimation under total variation smoothness. *Electronic Journal of Statistics*, 16(2), 3937–3972. <https://doi.org/10.1214/22-EJS2037>
- Li S., Chen Z., Zhu H., Wang C. D., & Wen W. (2023). Nearest-neighbor sampling based conditional independence testing. In *Proceedings of the AAAI Conference on Artificial Intelligence* (Vol. 37, pp. 8631–8639). AAAI Press.
- Li Y., Swersky K., & Zemel R. (2015). Generative moment matching networks. In *International Conference on Machine Learning* (pp. 1718–1727). Proceedings of Machine Learning Research (PMLR).

- Long J., Shelhamer E., & Darrell T. (2015). Fully convolutional networks for semantic segmentation. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition* (pp. 3431–3440). IEEE Computer Society.
- Muandet K., Fukumizu K., Sriperumbudur B., & Schölkopf B. (2017). Kernel mean embedding of distributions: A review and beyond. *Foundations and Trends® in Machine Learning*, 10(1-2), 1–141. <https://doi.org/10.1561/2200000060>
- Neykov M., Balakrishnan S., & Wasserman L. (2021). Minimax optimal conditional independence testing. *Annals of Statistics*, 49(4), 2151–2177. <https://doi.org/10.1214/20-AOS2030>
- Niu Z., Chakraborty A., Dukes O., & Katsevich E. (2024). Reconciling model-x and doubly robust approaches to conditional independence testing. *Annals of Statistics*, 52(3), 895–921. <https://doi.org/10.1214/24-AOS2372>
- Pan W., Tian Y., Wang X., & Zhang H. (2018). Ball divergence: Nonparametric two sample test. *Annals of Statistics*, 46(3), 1109. <https://doi.org/10.1214/17-AOS1579>
- Park J., & Muandet K. (2020). A measure-theoretic approach to kernel conditional mean embeddings. *Advances in Neural Information Processing Systems*, 33, 17104–17115. <https://doi.org/10.48550/arXiv.2006.05433>
- Pearl J. (2009). *Causality: Models, reasoning and inference*. Cambridge University Press.
- Pogodin R., Deka N., Li Y., Sutherland D. J., Veitch V., & Gretton A. (2022). ‘Efficient conditionally invariant representation learning’, arXiv, arXiv:2212.08645, preprint: not peer reviewed. <https://doi.org/10.48550/arXiv.2212.08645>
- Pogodin R., Schrab A., Li Y., Sutherland D. J., & Gretton A. (2024). ‘Practical kernel tests of conditional independence’, arXiv, arXiv:2402.13196, preprint: not peer reviewed. <https://doi.org/10.48550/arXiv.2402.13196>
- Ronneberger O., Fischer P., & Brox T. (2015). U-net: Convolutional networks for biomedical image segmentation. In *Medical Image Computing and Computer-Assisted Intervention–MICCAI 2015: 18th International Conference, Munich, Germany, October 5-9, 2015, Proceedings, Part III 18* (pp. 234–241). Springer.
- Scetbon M., Meunier L., & Romano Y. (2022). An asymptotic test for conditional independence using analytic kernel embeddings. In *International Conference on Machine Learning* (pp. 19328–19346). PMLR.
- Sen R., Suresh A. T., Shanmugam K., Dimakis A. G., & Shakkottai S. (2017). Model-powered conditional independence test. *Advances in Neural Information Processing Systems*, 30, 3039–3049. <https://doi.org/10.48550/arXiv.1709.06180>
- Shah R. D., & Peters J. (2020). The hardness of conditional independence testing and the generalised covariance measure. *Annals of Statistics*, 48(3), 1514–1538. <https://doi.org/10.1214/19-AOS1857>
- Sheng T., & Sriperumbudur B. K. (2023). On distance and kernel measures of conditional dependence. *Journal of Machine Learning Research*, 24(7), 1–16. <https://doi.org/10.48550/arXiv.1912.01103>
- Shi C., Xu T., Bergsma W., & Li L. (2021). Double generative adversarial networks for conditional independence testing. *Journal of Machine Learning Research*, 22(285), 1–32. <https://doi.org/10.48550/arXiv.2106.07229>
- Sriperumbudur B. K., Fukumizu K., & Lanckriet G. R. (2011). Universality, characteristic kernels and rkhs embedding of measures. *Journal of Machine Learning Research*, 12(70), 2389–2410. <https://doi.org/10.48550/arXiv.1003.0887>
- Su L., & White H. (2007). A consistent characteristic function-based test for conditional independence. *Journal of Econometrics*, 141(2), 807–834. <https://doi.org/10.1016/j.jeconom.2006.11.006>
- Su L., & White H. (2014). Testing conditional independence via empirical likelihood. *Journal of Econometrics*, 182(1), 27–44. <https://doi.org/10.1016/j.jeconom.2014.04.006>
- Szabó Z., & Sriperumbudur B. K. (2018). Characteristic and universal tensor product kernels. *Journal of Machine Learning Research*, 18(233), 1–29. <https://doi.org/10.48550/arXiv.1703.01950>
- Tang R., & Yang Y. (2023). Minimax rate of distribution estimation on unknown submanifolds under adversarial losses. *Annals of Statistics*, 51(3), 1282–1308. <https://doi.org/10.1214/23-AOS2291>
- Wang X., Pan W., Hu W., Tian Y., & Zhang H. (2015). Conditional distance correlation. *Journal of the American Statistical Association*, 110(512), 1726–1734. <https://doi.org/10.1080/01621459.2014.993081>
- Zafar A., Aamir M., Mohd Nawi N., Arshad A., Alruban A., Dutta A. K., & Almotairi S. (2022). A comparison of pooling methods for convolutional neural networks. *Applied Sciences*, 12(17), 8643. <https://doi.org/10.3390/app12178643>
- Zhang K., Peters J., Janzing D., & Schölkopf B. (2011). Kernel-based conditional independence test and application in causal discovery. In *Proceedings of the Twenty-Seventh Conference on Uncertainty in Artificial Intelligence* (pp. 804–813). Association for Uncertainty in Artificial Intelligence (AUAI) Press.
- Zhang X., Yao S., & Shao X. (2018). Conditional mean and quantile dependence testing in high dimension. *Annals of Statistics*, 46(1), 219–246. <https://doi.org/10.1214/17-AOS1548>
- Zhou X., Jiao Y., Liu J., & Huang J. (2022). A deep generative approach to conditional sampling. *Journal of the American Statistical Association*, 118(543), 1–12. <https://doi.org/10.1080/01621459.2021.1906684>
- Zhou Y., Shi C., Li L., & Yao Q. (2023). Testing for the Markov property in time series via deep conditional generative learning. *Journal of the Royal Statistical Society: Series B, Statistical Methodology*, 85(4), 1204–1222. <https://doi.org/10.1093/jrsssb/qkad064>